728780335 World Quant Brain Alpha Documentation

Quantitative Methods in Management (Trường Đại học Bách khoa Hà Nội)

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*Read this First * - Starter Pack

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2. Finance Basics
3. Quantitative Analysis
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Welcome to WorldQuant BRAIN

The objective this guide is to provide a basic overview of the Brain consultant program and simple bite-sized concepts to help you get started on alpha making. This is by no means an exhaustive list, but after completing this guide, you will gain sufficient knowledge to try your hands at simulating ideas. It will be a learning process, many have done this before, and so you can do it too regardless of your background. Good luck!

Research Consultant – An opportunity to work with WorldQuant

Finance Basics – The stock market and how it functions

Quantitative Analysis – And its relation to the BRAIN Platform

How to Use the BRAIN Platform – Fast Expression, Operator and Data Field

Common Financial Analysis Methods with Examples – Technical and Fundamental Analysis

Research Consultant

Become a Consultant with us

WorldQuant is a global quantitative asset management firm. Founded in 2007 by Igor Tulchinsky, our team of more than 850 employees worldwide develops and deploys

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countries and participate in WorldQuant's greater research efforts.

A Global Community

We seek to produce high-quality predictive signals (alphas) through our proprietary research platform so as to employ financial strategies focused on exploiting market inefficiencies. Our teams work collaboratively to drive the production of alphas and financial strategies. WorldQuant BRAIN® introduces its users to the intriguing world of quantitative finance in an interactive way through its simulation platform consisting of data sets and tools, performance dashboards and value-add measures.

80,000+ Users 3,500+ Consultants 120,000+ Data Fields

Benefits of being a Consultant

As a research consultant, you will have the following opportunities:

- Receive merit-based financial compensation. Earn up to $120 per day (Activity-based) and $25,000 per quarter (Performance-based)
- Opportunity to be considered for potential internships and full-time positions for eligible BRAIN consultants
- Membership within an elite community of global quants
- Access to create Alphas in more regions across Europe and Asia
- Access to 85,000+ datafields
- Access to special features: data-visualization, advanced webinar series, multi-simulations, SuperAlphas, BRAIN API documentation with Python etc.
- And much more.

Who is eligible?

No specific background required! Prior training in Science, Technology, Engineering, Mathematics (STEM) or any other related field that is highly analytical and quantitative will be useful.

Users need to score at least 10,000 points on the WorldQuant Challenge to be eligible for this role. At present, we are offering this opportunity exclusively to residents of:

- Mainland China
- Hong Kong
- Taiwan
- Kenya
- Korea
- Indonesia
- India

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- Vietnam
- Thailand
- USA

Read about the stories of our consultants :

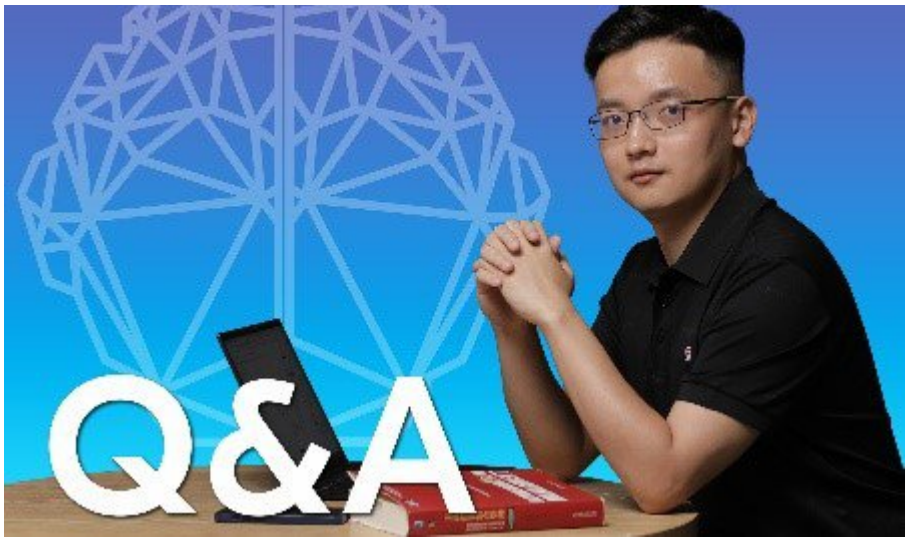


Azmi Fauzi – Indonesia



Aradhana Singh – India

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Zhuangzhuang Meng - China

Finance Basics

Many of our consultants started out from non-financial specializations, so don't be worried if you don't have prior financial knowledge. This section will brush you up with some basic know-how.

How the Stock Market works

The stock market refers to public markets that exist for issuing, buying, and selling stocks that trade on a stock exchange or over-the-counter. Stocks, also known as equities, represent fractional ownership in a company, and the stock market is a place where investors can buy and sell ownership of such investible assets [1].

Seeking Returns

Investors seek to profit from buying stocks by selling their stock for a profit if the stock price increases from their purchase price. For example, if an investor buys shares of a company's stock at $10 a share and the price of the stock subsequently rises to $15 a share, the investor can then realize a 50% profit on their investment by selling their shares [1]. Returns is defined by WorldQuant as the return on capital traded: .

$$AnnualReturn = AnnualizedPnL/HalfofCapitalTraded$$

It signifies the amount made or lost during the period and is expressed in %

Long a Stock

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On the other hand, taking a short position in a stock means borrowing an equity that you do not own, usually from your broker, then selling it, and hoping that it declines in value. When that happens, you can buy it back at a lower price than you have paid for it and return the borrowed shares to your broker.

Defining Volume

Volume is the amount of an asset or security that changes hands over some period of time, often over the course of a day. For instance, stock trading volume could refer to the number of shares of a security traded between its daily open and close. Trading volume, and changes to volume over the course of time, are important inputs for technical traders [2].

Defining what is Close / Open Price

The opening price is the price at which a security first trades upon the opening of an exchange on a trading day. The closing price is the price of the final trade before the close of the trading session. These prices are important because they are used to create traditional line stock charts, as well as when calculating moving averages and other technical indicators [3, 4].

Further Reading & References:

[1] Corporate Finance Institute. (2022, October 28). Stock market. https://corporatefinanceinstitute.com/resources/wealth-management/stock-market/

[2] What is volume of a stock, and why does it matter to investors? (2003, November 23). Investopedia. https://www.investopedia.com/terms/v/volume.asp#:~:text=Volume%20is%20the%20amount%20of,its%20daily%20open%20and%20close

[3] Close. (2003, November 18). Investopedia. https://www.investopedia.com/terms/c/close.asp

[4] Opening price: Definition, example, trading strategies. (2005, July 3). Investopedia. https://www.investopedia.com/terms/o/openingprice.asp#:~:text=The%20opening%20price%20is%20the,is%20its%20daily%20opening%20price

Quantitative Analysis

There are many methods to determine whether to long (buy) a stock or to short it.

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market data" to develop trading algorithms and computer models. The information generated by these computer models helps investors analyze investment opportunities and develop what they believe will be a successful trading strategy [5].

About the BRAIN Platform

Utilizing the quantitative analysis approach, the BRAIN platform is a web-based simulator of global financial markets that was created to explore Alpha research. It accepts an Alpha expression as input and plots its Profit and Loss (PnL) as output.



The input expression is evaluated for each financial instrument, every day over historical dates, and a portfolio is constructed accordingly. BRAIN platform invests in each financial instrument according to the value of the expression. It takes positions (either buying or short selling) and assigns weights to each instrument.

What is an Alpha?

An Alpha is an algorithm that transforms the input data (price-volume, news, fundamental, etc.) into a vector in which values are proportional to the positions and weights we want to hold in each instrument for each day.

Weights

Imagine market data being a matrix, with each row representing one date and each column representing one stock. For example, the matrix for close price data could look like this:

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20100104	30.95	214.01	626.75
20100105	30.96	214.38	623.99
20100106	30.77	210.97	608.26

Table: Closing stock prices of company A, B and C for 3 respective days.

The role of the Alpha expression is to transform the input matrix to an output vector of weights, with each weight corresponding to one of the stocks. The Alpha output vector, having weights as values corresponding to each instrument in the Universe, could look something like this:

Instruments	A	B	C
Alpha Weights	0.2	-0.5	0.3
Long or Short	Long	Short	Long

Table: Output vector to indicate the direction as well as sizing for company A, B and C.

Cumulative Profit & Loss (PnL)

Once we have got the weights of the stock from the Alpha expression, the next step is to get each day's profit and loss (PnL).

From the above table, I have weight_A = 0.2, weight_B = -0.5 and weight_C = 0.3. Now the amount of money I have got to invest is called the "booksize". Suppose my booksize is 100 USD. So I calculate the money I want to invest in each of the stocks:

money_A = 0.2 * 100 USD = 20 USD Long

money_B = -0.5 * 100 USD = 50 USD Short

money_C = 0.3 * 100 USD = 30 USD Long

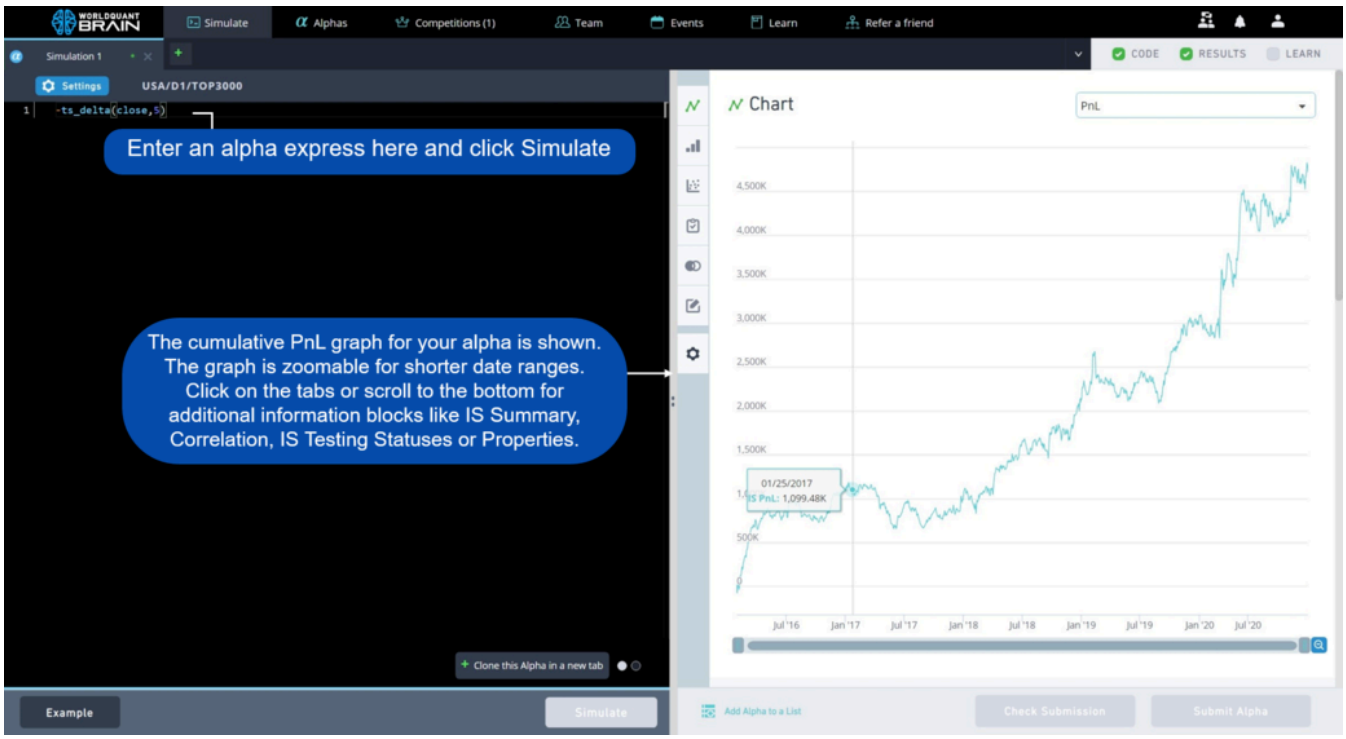
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has also changed, say from 100 USD to 105 USD. So, I have made a profit of 5 USD on that day.

Now I again calculate the Alpha values for the stocks, and again calculate weights, and again trade 100 USD worth of portfolio. [Note: In BRAIN platform, we use constant book size for all the days, regardless of whether your portfolio makes money or loses money.]

This is repeated for each day in the simulation period to calculate and plot the cumulative PnL.



Reducing Risk and Volatility

A good Alpha would ideally have consistently increasing PnL, high Annual Return, and more importantly, few fluctuations in the cumulative profit graph. If the standard deviation is low, there would be lesser fluctuations in the graph. If the graph shows high fluctuations/volatility, despite the returns being high, the Alpha will not be deemed good enough.

WorldQuant aims to develop equity long-short market neutral alphas that have low volatility and risk. Such investments are attractive because they are expected to produce substantially better risk-adjusted returns than long-only portfolios. Equity long-short market neutral strategy is used commonly by hedge funds, with the goal of minimizing exposure to the market and profit from the changes in the spread between two stocks.

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https://www.investopedia.com/articles/investing/041414/simple-overview-quantitative-analysis.asp

How to Use the Brain Platform

Enough theory for now, let's delve into how to write your first alpha on the BRAIN platform.

No Coding Experience Required

To all the non-coders, the good news of using BRAIN is that no prior coding experience is required.

BRAIN's Way of Coding

BRAIN uses the fast expression language that consists of two main elements: Data fields and Operators.

Can I use Python / R / MATLAB etc. for Alphas?

One of our user has asked us: Are there plans to allow users to utilize Python/MATLAB/R to interact with the BRAIN API, to analyze datasets and submit the alpha vectors?

Our reply is that: the BRAIN platform is currently available for use only with Fast Expressions language. Regarding APIs, we currently do not prohibit programmatic access to BRAIN platform when API communication is conducted with low intensity.

Data Fields, Datasets

Data fields refer to a named collection of data, for example 'open price' or 'close price'.

Datasets are a collection of data fields. For example, 'open price' and 'close price' can be found in the price volume dataset here. Most users typically start out with price volume and fundamental datasets.

Operators

An Operator is a set of mathematical or statistical techniques required to implement your Alpha ideas, for example mathematical operators: + - / * or cross-sectional operators such as 'rank'. Read Learn/Operators for more details.

Here are some common examples of alphas build using data fields and operators:

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1/close	Use inverse of daily close price as stock weights. More allocation of capital on the stocks with lower daily close prices.
volume/adv20	Use relative daily volume to the average volume in the past 20 days as stock weights.
ts_corr(close, open, 10)	Use correlation between daily close and open prices in the past 10 days as stock weights.
rank(adv20)	Use rank of average daily volume in the past 20 days (adv20) as stock weights.
ts_delta(close, 5)	Use difference between daily close and close on the date 5 days earlier as stock weights.

You can try out some sample alphas by clicking on the Example button (bottom left corner) on the simulator page. Leverage the hint and test out a few simulations! Click here to try now: Simulate Page

Methods of Analyzing the Stock Market

You might ask us, how do you come up with ideas for new alphas? This section will take you through two alpha ideas utilizing technical analysis and fundamental analysis, and explain the thought process behind them.

Technical Analysis

Technical analysis is a trading discipline employed to evaluate investments and identify trading opportunities by analyzing statistical trends gathered from trading activity, such as price movement and volume. Technical analysts believe past trading activity and price changes of a security can be valuable indicators of the security's future price movements.

Across the industry, there are hundreds of patterns and signals that have been developed by researchers to support technical analysis trading. These include trend lines, channels, moving averages, and momentum indicators [6].

Volume as an indicator

If a company's stock has high volume, it means that many people are buying and selling the stock. Suppose our hypothesis is that the company with more shares traded is more desirable than another company with low volume. We will then allocate more weight to the company with higher volume.

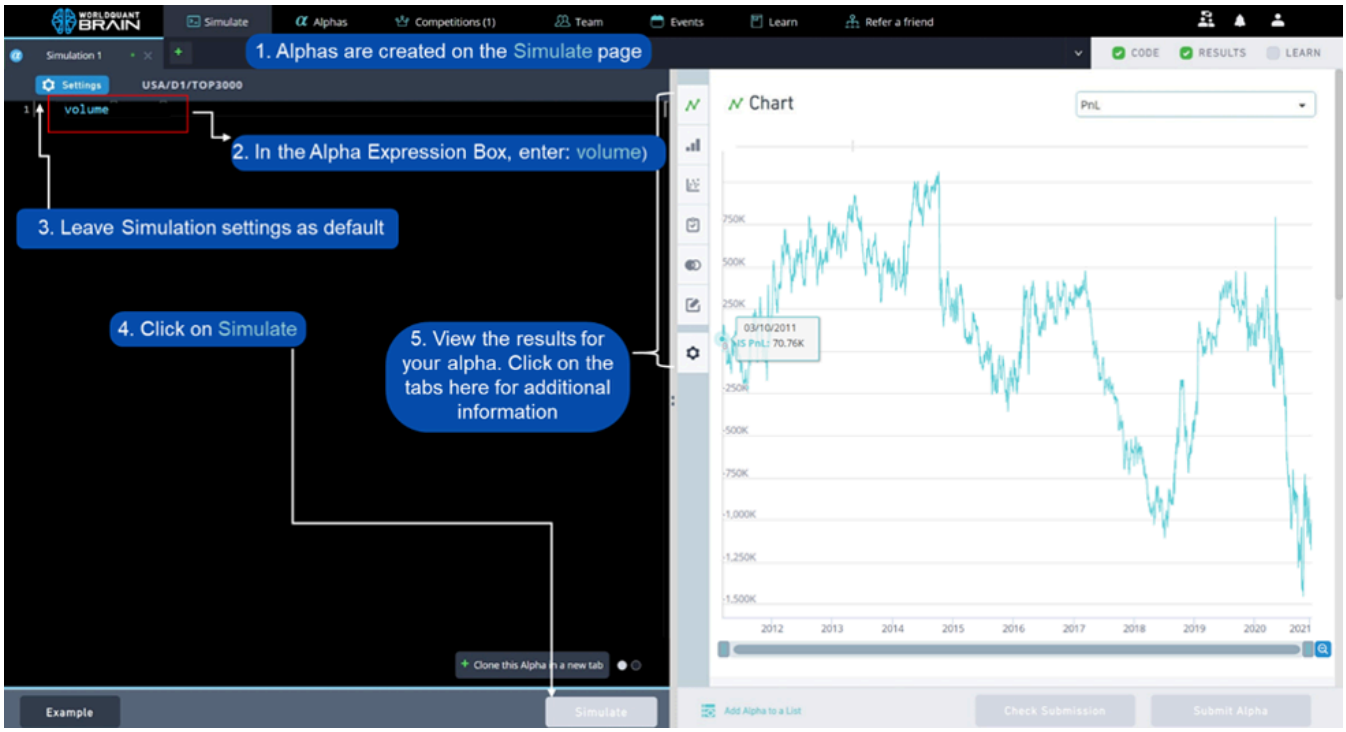
One way to express this idea is through the alpha expression here:

```
volume
```

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Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	1	1	5	Market	On	Off	Verify



Fundamental Analysis

Fundamental analysis is a method of measuring a security's intrinsic value by examining related economic and financial factors. Fundamental analysts study anything that can affect a security's value, from macroeconomic factors such as the state of the economy and industry conditions to microeconomic factors like the effectiveness of the company's management.

Analysts could compare the company's growth rates to the industry and sector that it operates in, along with the other information provided, to see if the company is valued correctly [7].

Inventory Turnover

Financial ratios are ratios of fundamental data which give insights into the health and investment decisions of the company. One common financial ratio is the inventory turnover. It is a type of activity ratio that measures how quickly a company sells through and replaces its inventory. It is calculated as:

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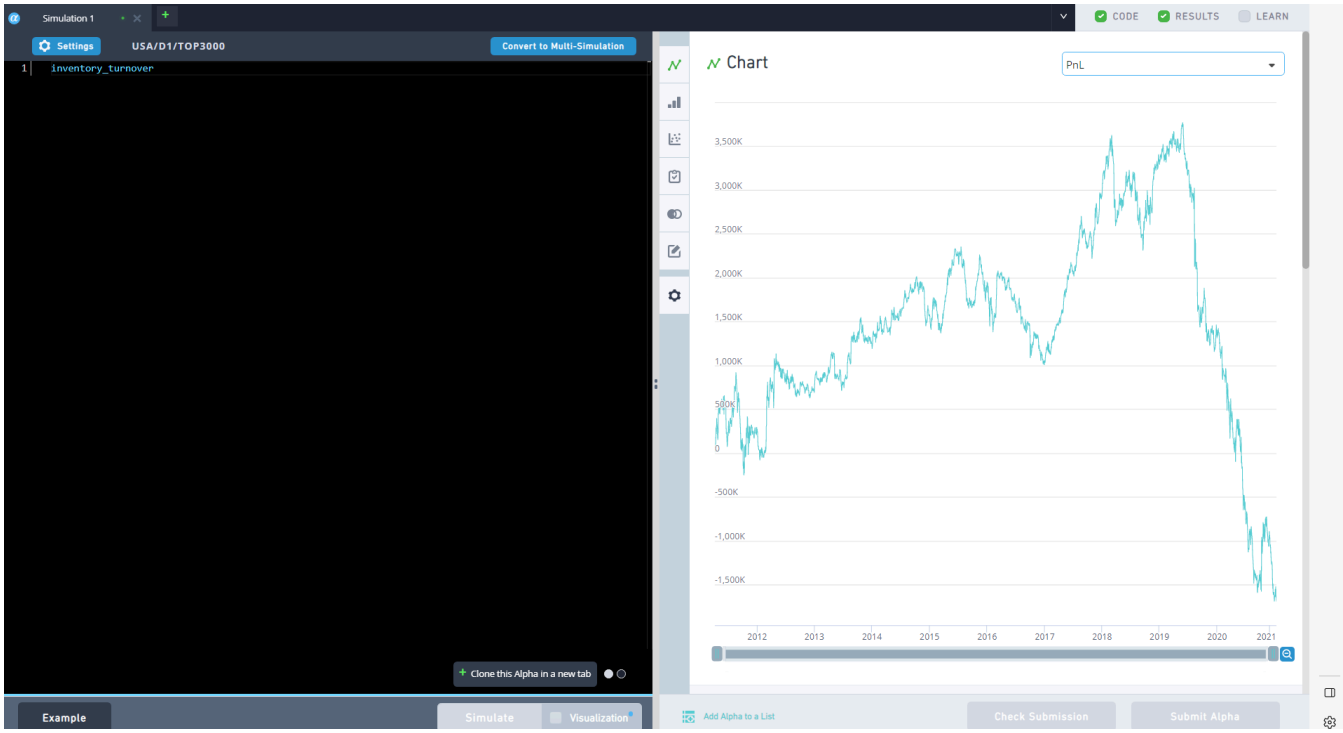
The hypothesis is that a stock with higher inventory turnover ratio will have a better performance and thus be allocated more weight.
The alpha expression is as such:

```
1 inventory_turnover
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	3	1	5	Market	On	Off	Verify



Time Series & Cross Sectional

Additionally, there are two frequently used operator categories: time series and cross sectional.

Time series analysis can be useful to see how a given variable changes over time. Suppose you wanted to analyze a time series of daily closing stock prices for a given stock over a period of one year. You could obtain a list of all the closing prices for the stock from each day for the past year and analyze the time series data with technical analysis

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company to its industry peers. Cross sectional analysis may focus on a single company for head-to-head analysis with its biggest competitors or it may approach it from an industry-wide lens to identify companies with a particular strength. Essentially, cross-sectional analysis shows an investor which company is best given the metrics you care about [9].

We hope you've enjoyed this guide! To get started, you may click on the Example Button (bottom left corner) on the Simulate Page. There you will find some sample alphas with hints to improve it.

If you have any research related questions, you can check out our Community forum and post your questions there.

Further Reading & References:

[6] Technical analysis: What it is and how to use it in investing. (2003, November 24). Investopedia. https://www.investopedia.com/terms/t/technicalanalysis.asp

[7] Fundamental analysis: Principles, types, and how to use it. (2003, November 23). Investopedia. https://www.investopedia.com/terms/f/fundamentalanalysis.asp

[8] Time series definition. (2006, March 12). Investopedia. https://www.investopedia.com/terms/t/timeseries.asp

[9] What is cross sectional analysis and how does it work? (2007, May 21). Investopedia. https://www.investopedia.com/terms/c/cross_sectional_analysis.asp

Next: Introduction to Alphas

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Documentation Next: Introduction to BRAIN Expression Language

Introduction to Alphas

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1. Alphas
2. Alpha Lifecycle
3. Weights
4. Assigning weights
5. Positive and Negative weights
6. BRAIN platform

BRAIN platform is a web-based tool for backtesting trading ideas. An Alpha is a concrete trading idea that can be simulated historically.

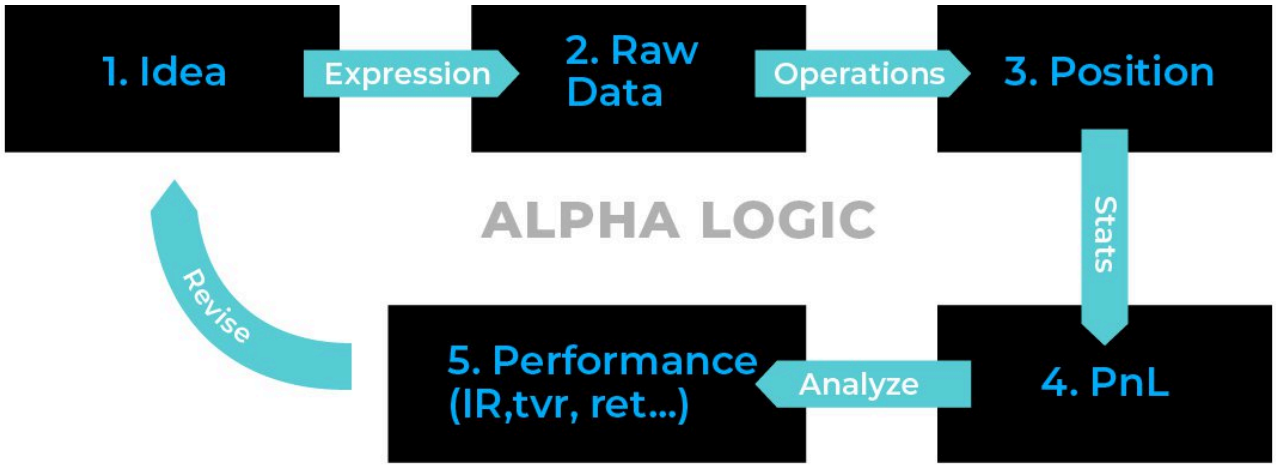
Alphas

In BRAIN platform, an 'Alpha' refers to a mathematical model or strategy, written as an expression, which places different bets (weights) on different instruments (stocks), and is expected to be profitable in the long run. After a user enters an Alpha expression that consists of data, operators and constants, the input code is evaluated for each instrument to construct a portfolio. Then BRAIN platform makes investments in each instrument for a one-day period in proportion to the values of the expression. The process repeats each day.

Alpha Lifecycle

The flow chart below shows the lifecycle of an Alpha:

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First, one might peruse blogs, journals and research papers on the internet to come up with an idea. The Alpha expression is entered in BRAIN platform and operations (like truncation,neutralization, decay) are performed on the raw Alpha. BRAIN platform makes investments (goes long or short) for all the instrumentsof the universe chosen in the Settings panel and the PnL is simulated. Then the performance is calculated (Sharpe, Turnover, Returns) as seen in the Simulation Results page. And if the Alpha is not deemed worthy, the Alpha idea is revised. Else, it enters production.

Weights

In simple terms, BRAIN platform uses an Alpha to create a vector of weights, with each weight corresponding to one of the stocks in the selected universe. These weights may or may not be market neutralized, as per your neutralization setting (by market, industry, sub-industry or none). This creates a portfolio for each day in the simulation period, which can then be used to calculate that day's Profit and Loss (PnL).

Assigning weights

Suppose in the Expression box, you type in 1/close, and set the simulation settings as follows: Region = US, Universe = TOP3000, Delay = 1, Decay = 0, Neutralization = None, Truncation = 0. Now, once you hit "Simulate" button, then for each day in the Simulation Duration (5 years), the simulator does the following:
It calculates 1/close (using the closing price for the previous day), for each instrument in the basket "US: TOP3000" (i.e. top 3000 stocks in the US, by market capitalization). This creates a vector of 3000 values (one for each stock). This vector is then normalized, i.e. divided by the sum of its values(so that all the values sum up to 1). This creates a vector

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each day in the simulation period, and at the end of each day the total profit or loss made by our portfolio is calculated.

Positive and Negative weights

BRAIN platform assigns positive weights to indicate long positions in stocks, and negative weights to indicate short positions in stocks. The greater the magnitude of the weight on a stock, the larger the long or short position taken on it.

It is easy to invest $100 in stock, but negative positions (shorts) are common too. E.g. one can get $100 now by shorting a stock (i.e. investing -$100), which must be bought back later. PnL would be the opposite of $100 invested, as seen in the table below. Negative weights are called short positions and positive weights are called long positions. Typically investors take short positions when they expect stock price to decrease and long positions (i.e. buying stocks) when they expect price to increase. Please refer to Investopedia for more details on short selling.

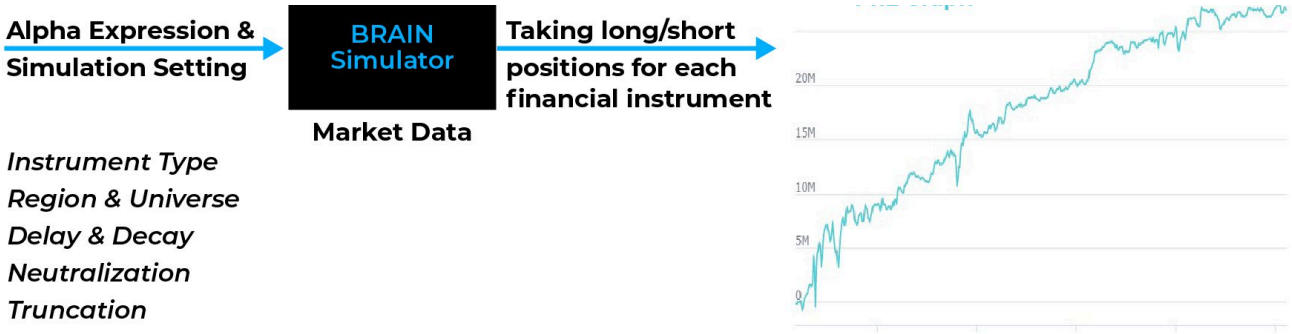
The below table gives an example of payoffs from a $100 short and a $100 long position, for a 1% price change in each direction. For simplicity, it does not account for dividends, margin and financing costs.

Position	Name	PnL if stock price rises 1%	PnL if stock price falls 1%
$100	Long	$1	-$1
-$100	Short	-$1	$1

BRAIN platform

BRAIN platform is a web-based simulator of global financial markets that was created to explore Alpha research. It accepts an Alpha expression as input and plots its Profit and Loss (PnL) as output. The input expression is evaluated for each financial instrument, every day over historical dates, and a portfolio is constructed accordingly. BRAIN platform invests in each financial instrument according to the value of the expression. It takes positions (either buying or short selling) and assigns weights to each instrument. The weights are then scaled to book size (amount of money invested), based on which a PnL graph is plotted. These weights are not constant; they change over time based on current information and the history of the changes of some variables (such as prices, volumes, etc.).

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Prev: *Read this First * - Starter Pack

Next: Introduction to BRAIN Expression Language

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Documentation Next: Intermediate Pack - Understand Results [1/3]

Introduction to BRAIN Expression Language

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1. What is Fast Expression?
2. Characteristics of Fast Expression
   - Data fields
   - Operators
3. Further Knowledge of Fast Expression

What is Fast Expression?

“Fast expression” is a proprietary programming language used by WorldQuant BRAIN that is designed to make it easier to write and test financial models. The language can be thought as a form of pseudo code, which uses natural language and simple programming constructs to convey the logic of the algorithm.

The goal of using “Fast expression” on BRAIN is to provide a clear and concise way to express complex ideas and algorithms that can be easily understood by other developers and researchers. By abstracting away the details of the underlying implementation, it can allow BRAIN users to focus on the high-level logic of their algorithms, rather than getting bogged down in the implementation details.

Characteristics of Fast Expression

Just like how an English sentence consists of a subject, verb and object; Fast expression can include data fields, operators and numerical values.

Data fields

Data fields refer to a named collection of data, for example 'open price' or 'close price'

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Company Fundamental Data for Equity	1,575
US News Data	644
…	…
Relationship Data	82
Options Analytics	74
Price Volume Data for Equity	72

A dataset containing price, volume, close, open price for each stock

Field	Description	Type
adv20	Average daily volume in past 20 days	Matrix
close	Daily close price	Matrix
returns	Daily returns	Matrix
volume	Daily volume	Matrix

Operators

Operators refer to a set of mathematical techniques required to implement your Alpha ideas.

1. Arithmetic Operators
2. Vector Operators
3. Logical Operators
4. Transformational Operators
5. Cross Sectional Operators
6. Group Operators
7. Time Series Operators

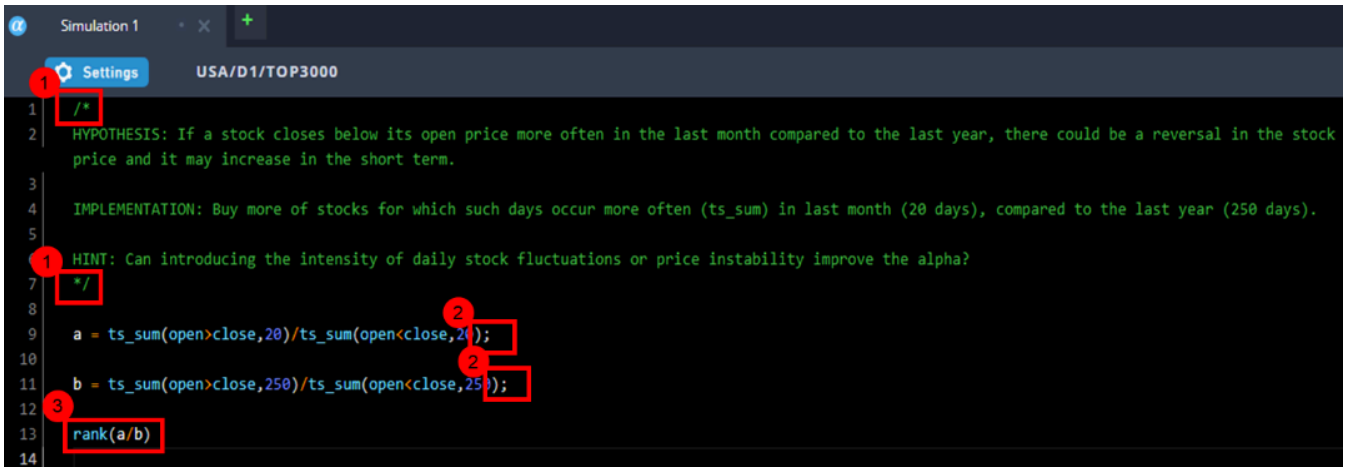
Arithmetic Operator	Description
+ , - , * , / , ^	Arithmetic operators: add, subtract, multiply, divide, power
log(x)	Natural logarithm

Cross Series Operator	Description
rank(x, n)	Ranks the input among all the instruments and returns an equally distributed number between 0.0 and 1.0

Time Series Operator	Description
ts_rank(x, n)	Rank the values of x for each instrument over the past n days, then return the rank of the current value. The rank value is between 0.0 and 1.0

Further Knowledge of Fast Expression

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- ***/**** helps to create block comments that span multiple lines of text, while ****/*** denotes the end of the comment. Comments consist of explanatory text to help understand what the code does. [1]
- ***;*** (semicolon) acts as a semicolon in a sentence, separating the end of one sentence from the beginning of another sentence. For the last line of the code (line 13) ; is not needed. [2]
- The last sentence of the entire expression is the alpha expression that the BRAIN simulator use to calculate the positions to take in each stock. [3]

Lastly, Fast expression does not have classes, objects, pointers, or functions.

In summary, Fast expression provides a clear and concise way for users to express complex ideas and algorithms. Don't worry if you're not familiar with Fast expression yet. With a bit of practice, we believe you'll pick it up in no time!

Prev: Introduction to Alphas

Next: Intermediate Pack - Understand Results [1/3]

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Documentation Next: Intermediate Pack - Improve your Alpha [2/3]

Intermediate Pack - Understand Results [1/3]

Table of Contents

1. Understanding Your Results
2. Passing IS Stage and Troubleshooting
3. Common Error Messages

This Intermediate guide aims to further your understanding of the alphas you have simulated. The documentation will provide you with an in-depth understanding of commonly used operators and get you up to speed to improve your ability to create a high-performing alpha.

Understanding Your Results

Cumulative PnL Chart



If you've followed the examples in the Starter pack, chances are you've ended up with the first 2 graphs. What both graphs have in common is that they have multiple periods of significant losses, producing a graph with high fluctuations. This means that your simulated portfolio could lose a large percentage of its value in one day, and that wouldn't be ideal. Rather, a good alpha should produce a steadily rising PnL chart (3rd

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In-sample simulation uses data over a 5-year timeframe, and tests out how well your alpha performs in the historical period. After the simulation, you will see the IS Summary row with 6 metrics: Sharpe, Turnover, Fitness, Returns, Drawdown, and Margin.



Sharpe

This ratio measures the excess return (or risk premium) per unit of deviation of returns of an Alpha. It takes the mean of the PnL divided by the standard deviation of the PnL. The higher the Sharpe Ratio or Information Ratio (IR), the more consistent the Alpha's returns are potentially likely to be, and consistency is an ideal trait. The passing requirement for Sharpe on the BRAIN platform is to be above 1.25.

$$Sharpe = \sqrt{252} * \left(\frac{Mean(PnL)}{Stdev(PnL)}\right)$$

Turnover

Turnover of an Alpha is metric that measures the simulated daily trading activity, i.e., how often the Alpha trades. It can be defined as the ratio of value traded to book size. The higher the turnover, the more often a trade occurs. Since trading incurs transaction costs, reducing turnover is generally an ideal trait. The passing requirement for turnover on the BRAIN platform is to be between 1% and 70%.

$$Turnover = \frac{DollarTradingValue}{Booksize}$$

Fitness

Fitness of an Alpha is a function of Returns, Turnover & Sharpe. Fitness is defined as:

$$Fitness = Sharpe * \sqrt{\frac{abs(Returns)}{max(Turnover, 0.125)}}$$

Good Alphas generally have high fitness. You can seek to improve the performance of your Alphas by increasing Sharpe (or returns) and reducing turnover. The passing

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Returns is the amount made or lost by the Alpha during a defined period and is expressed in percentages. BRAIN defines returns as:

$$AnnualReturn = \frac{AnnualizedPnL}{0.5 * BookSize}$$

Drawdown

Drawdown of an Alpha is the largest reduction in simulated PnL during a given period, expressed as a percentage. It is calculated as follows:

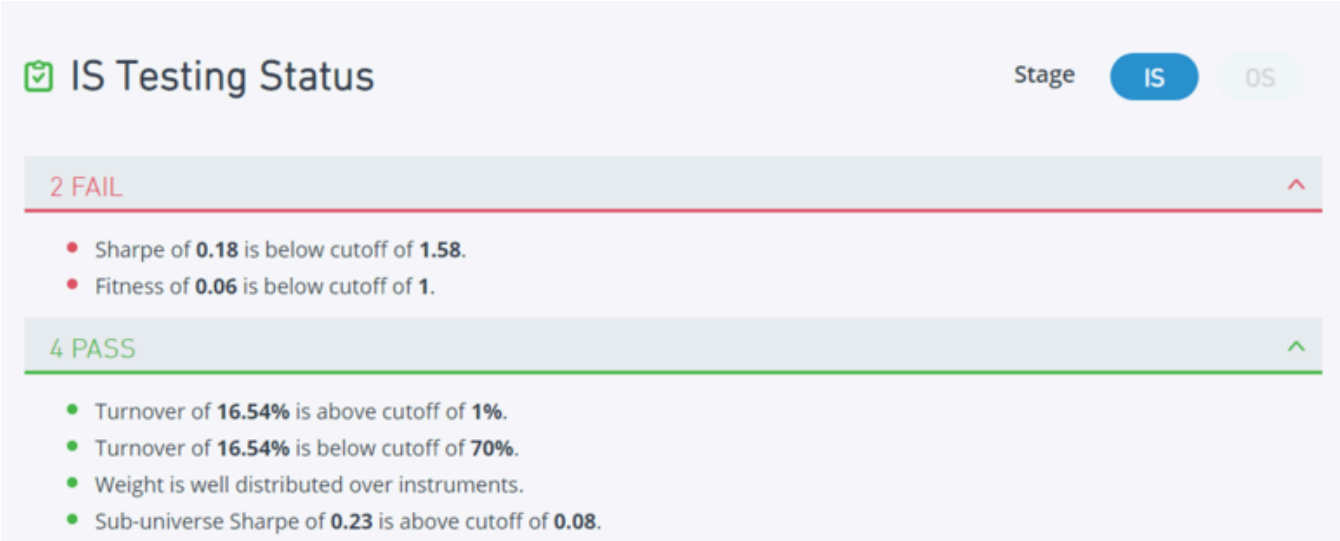
$$Drawdown = \frac{DollarAmountOfLargestPeakToTroughGapInPnL}{0.5 * BookSize}$$

Margin

Margin is the simulated profit per dollar traded of an Alpha; calculated as:

$$Margin = \frac{PnL}{TotalDollarsTraded}$$

Passing IS Stage and Troubleshooting



- One of the most common challenges users face is Low Sharpe, and users commonly see that their Sharpe ratio is below the specified cutoff. How do you get a higher Sharpe? We suggest that you can either increase you Alpha return or reduce your volatility. Read more here.
- Another challenge is the weight test that measures the capital concentration in each

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Read more here.

- Another difficulty is that the Sub-universe Sharpe is not above cutoff. This means that the Sharpe in the sub-universe must be higher than at least one threshold. There are 2 thresholds that scale down Sharpe with sub-universe size.

$$Threshold\ 1 = 0.75 * \sqrt{250} * \max(0.065, \sqrt{\frac{Univers\ e_{size}}{Subunivers\ e_{size}}} * N$$

$$Threshold\ 2 = 0.75 * \frac{Original\ Sharpe}{\frac{Universe_size}{Subuniverse_size}}$$

Thus, you can try to improve the Sub-Universe Sharpe by increasing the Universe of instruments (i.e. selecting Top3000).

Common Error Messages

Syntax error in expression.

Check your spelling of the data fields and operators and ensure that your expression is logical. The tokens (operators and keywords) allowed in your Alpha expression can be found in the Available Market Data and Available Operators pages. Alpha expressions also accept integers and floating point numbers.

Incompatible unit for input at index 0, expected "Unit[]", found "Unit[CSPrice:1]"

Unit warnings are provided for reference in simple cases and do not prevent submission. Usually, this warning appears when data fields having two different units are added or multiplied. E.g. if you add "close" to "cap". "close" has units of price but "cap" has units of price*shares. You can safely ignore these warnings if you're sure the alpha correctly handles data units.

Prev: Introduction to BRAIN Expression Language Next: Intermediate Pack - Improve your Alpha [2/3]

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Intermediate Pack - Improve your Alpha [2/3]

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1. Use Different Operators
2. Change Simulation Settings

Use Different Operators

Remember the cross-sectional operators and time-series operators from the Starter pack? You can use them here. Secondly, you can try out different data fields. We recommend exploring the price volume dataset, model dataset and fundamental dataset. Lastly, you can tinker with different simulation settings. These tips should help improve the performance of your second Alpha.

Divide (/)

We can divide datafields with other datafields.

Imagine market data being a matrix, with each row representing one date and each column representing one stock. For example, the matrix for close price data of stocks in universe US TOP3000 would look like this:

Dates	Close Price of A	Close Price of B	Close Price of C	...
2010-01-04	30.95	25.46	214.01	...
2010-01-05	30.96	25.65	214.38	...
2010-01-06	30.77	25.59	210.97	...
...	...	...	...	...

And the matrix for open data of above stocks would look like this:

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2010-01-04	30.74	25.78	213.82	...
2010-01-05	30.99	25.52	214.10	...
2010-01-06	30.83	25.64	211.35	...
...	...	...	...	...

Say you enter an Alpha expression like ***close/open*** in the Simulate page found in the Alphas dropdown tab. When you click Simulate, the BRAIN platform will evaluate the Alpha expression against the matrix of market data for each date and each stock.

Rank(x)

Description: the Rank operator ranks the value of the input data x for the given stock among all instruments, and returns float numbers equally distributed between 0.0 and 1.0

Alpha expression: ***rank(sales/assets)***. If company B has a higher asset turnover ratio (sales/assets) than company A, stock B may outperform stock A. The rank operator helps to limit the extreme values of that ratio.

For example:

Stock	A	B	C	D	E
Alpha Expression ***sales/assets***	1	2	3	20	100

The numbers imply that if you have $126, you must use $100 to go long stock E (~80% of your total capital). So, your strategy would depend crucially on how the last stock performs. But, isn't that too risky? Applying the rank function to the alpha expression ***rank(sales/assets)***, you get:

Stock	A	B	C	D	E
Alpha Expression ***rank(sales/assets)***	0	0.25	0.5	0.75	1

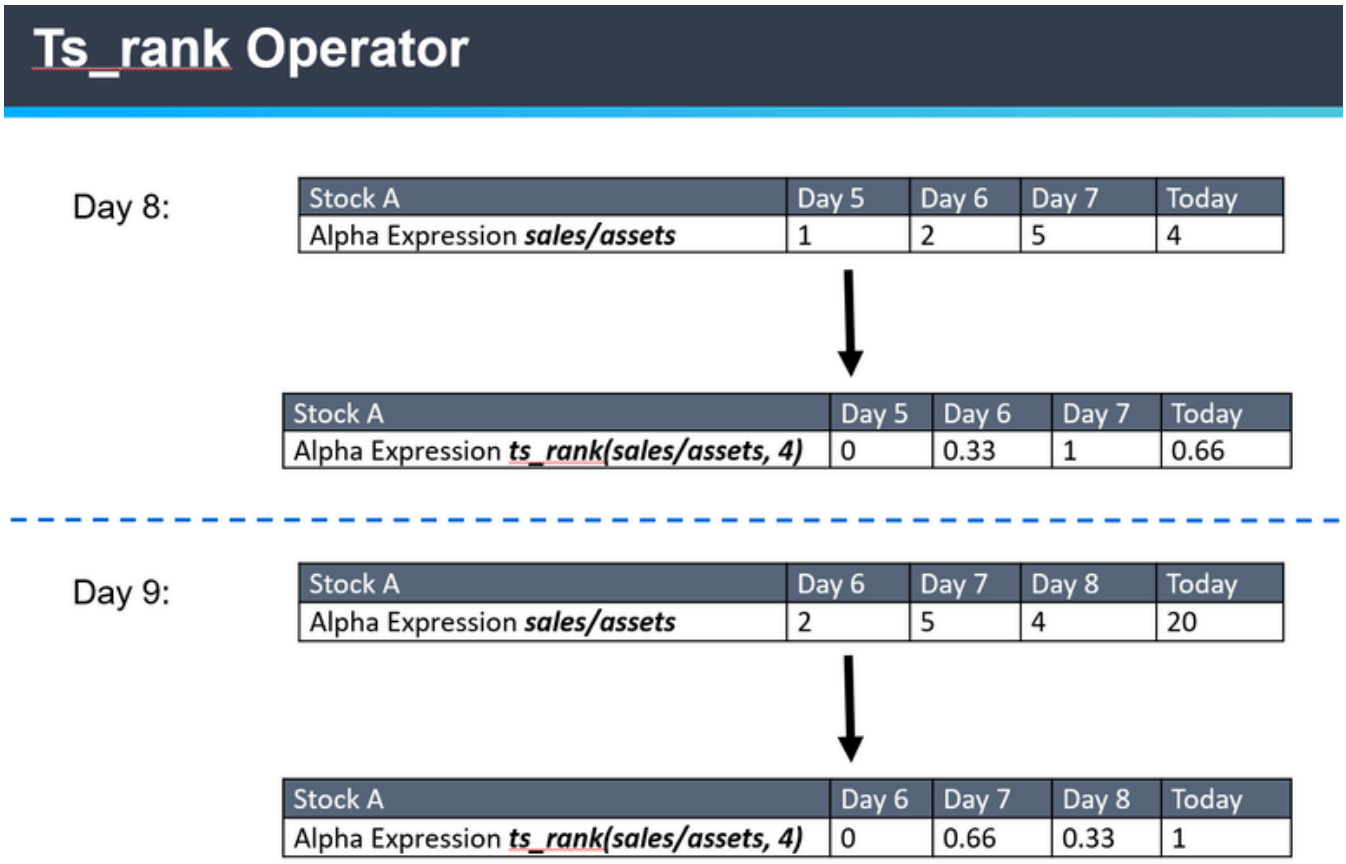
This time you see that the stock with the largest weight occupies only 40% of your portfolio.

Ts_rank & Ts_delta Operator

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Description: ranks the values of x for each instrument over the past d days, then return the rank of the current value. The rank value is equally distributed between 0.0 and 1.0	Description: Returns the difference of x and the x value d days ago
Alpha expression: ***ts_rank(sales/assets,60).*** If company A has a higher asset turnover ratio (sales/assets) compared to its historical data, the company may be growing and its stock price might follow. The ts_rank operator compares the historical value of the ratio over the past 60 days and assigns 1 if it is at its highest, and 0 if it is at its lowest.	Alpha expression: ***-ts_delta(close,2)***. If the stock price of a company has increased over the last 2 days, it may decrease in the future. T ts_delta operator takes the difference of the close price today and the close price 2 days ago.

Visual Illustration of Ts_rank Operator:

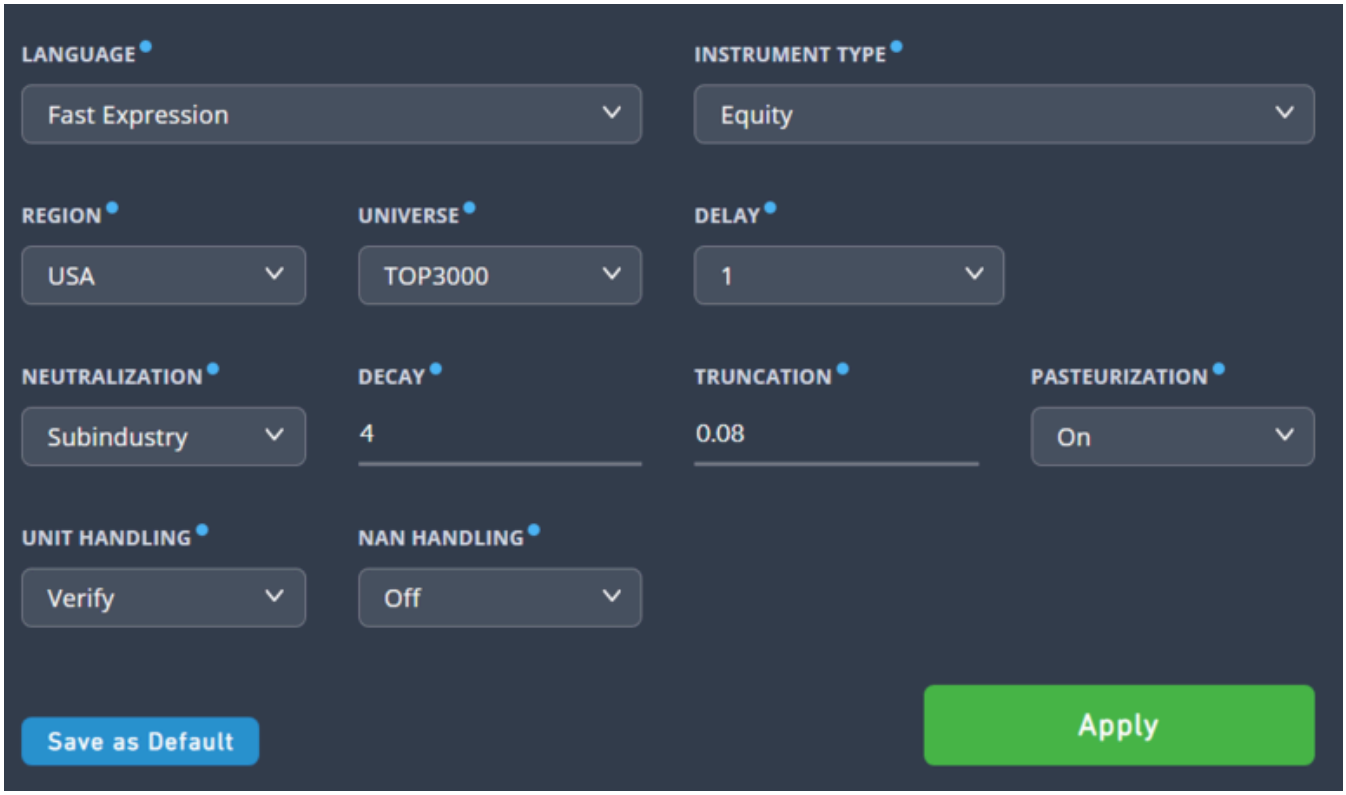


Change Simulation Settings

In your first Alpha simulation, you left the simulation settings on default. Changing certain

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later guide.



Region

Region refers to the market in which the Alpha will simulate trades, for example, the U.S. equity market or Chinese equity market.

Universe

Universe is a set of trading instruments ranked by their liquidity. For example, “US: TOP3000” represents the top 3,000 most liquid stocks in the U.S. market.

Decay

Decay is used for averaging the Alpha signal within a specified time window. The settings perform linear decay on the Alpha. Tip: Decay can be used to reduce turnover, but decay values that are too large will attenuate the signal.

$$Decay_linear(x,n) = \frac{x[date]*n + x[date\ -1)*(n-1) + \cdots + x[date-n-1]}{n+(n-1)+\cdots+1}$$

Truncation

Truncation sets the maximum weight for each stock in the overall portfolio. It aims to

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Neutralization

Market risks and industry specific risks are prevalent risks within equities. However, these risks can be reduced by creating long-short neutral portfolios using a concept called neutralization. After neutralizing the portfolios to market or industry specific groups, no net position is taken with respect to that group, i.e. allotting the same amount of dollars in long (buying) and short (selling) positions. That way, you are less exposed to risk, whether the entire market goes up or down.

When Neutralization = "Market/Industry/Sub-industry" it does the following operation: *Alpha = Alpha – mean(Alpha)* where Alpha is the vector of weights.

Before Neutralization		After Neutralization	
alpha_A	alpha_B	new_alpha_A	new_alpha_B
2	4	-1	1

If the hypothetical booksize is 20 million, we would end up investing $10 million in long positions and $10 million in short positions. Thus, no net position is taken with respect to the market. In other words, the long exposure cancels out the short exposure completely, making this hypothetical strategy market neutral.

The three different neutralization methods determine which groups are used for neutralizing Alpha values. The correct choice of neutralization depends on the logic or formula used by the Alpha. The results should indicate which neutralization will be most effective.

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Documentation Next: WorldQuant Challenge

10 Steps to Start on BRAIN platform

10 Steps to Start on BRAIN platform

1. Watch the “Introduction to Alphas” training series
2. Read the starter pack series under “Discover BRAIN” section
3. Simulate 3 simple formulas using the data field: close and the operator: rank
4. Understand the simulation settings, starting with delay and neutralization .
5. Click the Example button on the bottom left of the Simulate page for examples. Try to improve these expressions using the Hint and then submit
6. Understand the Simulation Results and submission tests.
7. Simulate Alphas discussed in the "Create Alphas" sections Run your First Alpha and Another Sample Alpha. Pay attention to the settings.
8. Check the Events page to attend training webinars.
9. Read the BRAIN TIPS series and the Beginner posts in FAQs/ Forums
10. Try to create Alphas using available Price volume data fields or and, Operators

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Documentation

WorldQuant Challenge

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1. Overview
2. Scoring criteria
   - Summary
   - Details

Overview

The WorldQuant Challenge is a perpetual, online, solo competition. Users can submit Alphas to improve their scores and ranking.

Individuals who score 10,000 points may be eligible to receive an invitation for the research consultant opportunity, subject to other criteria(e.g. if they are residents in countries where the BRAIN consultant program is offered). Users who make it to Gold and Silver levels will have access to special training sessions and videos through the Events page.

New users are automatically enrolled into the challenge. The Leaderboard ranks all eligible users and can be filtered by country, university and/or city.

Scoring criteria

Summary

1. Your score is based on the quantity and quality (performance in the 5 year in-sample period) of Alphas that you submit on the platform
2. Your score also depends on quantity and quality of Alphas submitted by other users that day

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5. There are no negative points. Your score cannot decrease
6. Scores refresh once every day at 3 AM EST
7. Participants with the same score will have the same rank
8. You can reach 3 levels in WorldQuant Challenge:
    1. Bronze (score > 1,000)
    2. Silver (score > 5,000)
    3. Gold (score > 10,000)

Details

Each day, all Alphas submitted by a user accumulated and two factors are calculated:

Quantity Factor: Larger the number of Alphas you submit during a day. Larger the factor, higher your score

Quality factor: Quality factor is calculated as an average of the quality factor of all Alphas submitted during the day. Larger the factor, higher your score. It depends on the following settings and results in the in-sample period:

- Universe (smaller universes get more score)
- SelfCorrelation (the lesser the better)
- Fitness (the higher the better)
- Delay (D1 Alphas contribute more to the score than D0 Alphas)

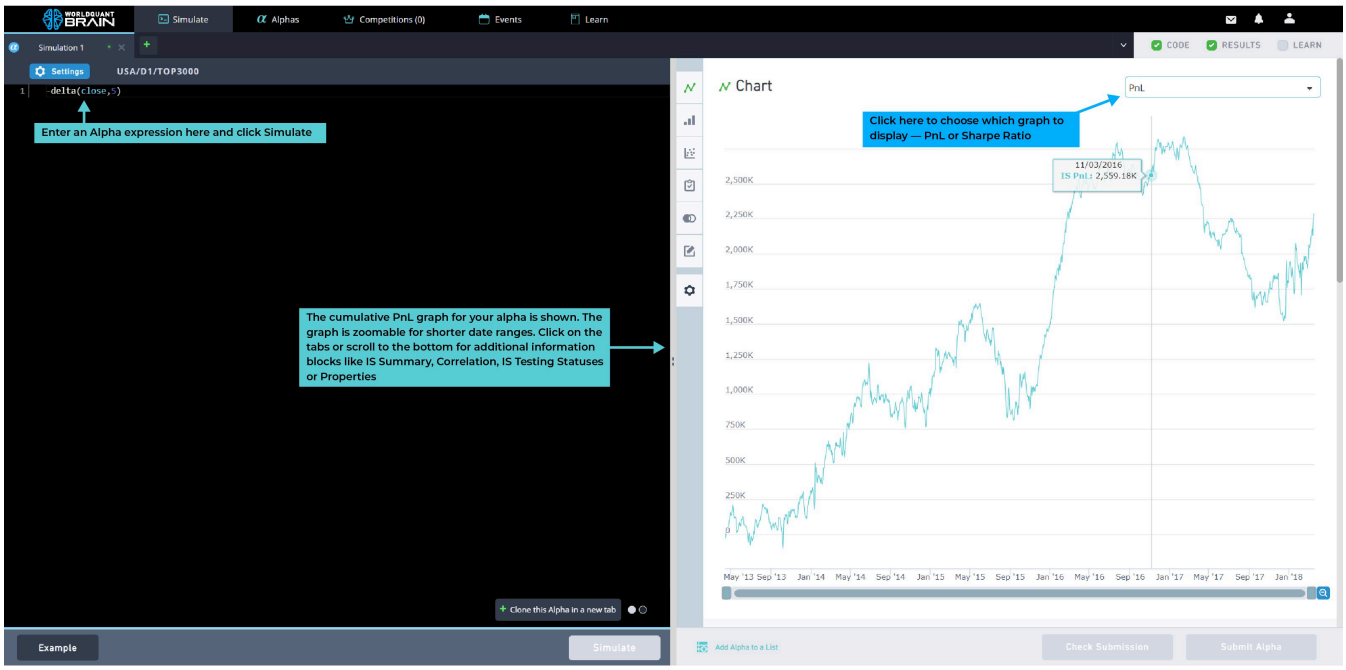
Both factors are then normalized across all the users who submitted at least one Alpha on that particular day. Your final daily score is then function of normalized Quantity and Quality Factors. The daily score is capped at 2,000 points.

Prev: 10 Steps to Start on BRAIN platform

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Simulate your first Alpha

Alphas are created and simulated on the Simulate page in the Alphas dropdown tab. To run your first simulation, click on the gear icon at the top right-hand side corner. This will open the settings panel. Here, select “US: TOP3000” for Region and Universe, “Subindustry” for Neutralization and apply your settings. Make sure both Code and Result are ticked by clicking on them. In the Alpha expression text box, enter **-Delta(close, 5)** for now and click on "Simulate". The Simulation Result page will show a graph for Cumulative Profit. This graph can be zoomed in to plot area for shorter time periods (1 month or 1 year).



The display consists of 2 graphs, one for PnL vs. Time and the other for Sharpe Ratio vs. Time.

In the Stats tab, a good Alpha would have consistently increasing PnL and high Annual Return, Sharpe Ratio, % Profitable Days and Profit per Dollar Traded. It should have low Drawdown and Turnover. And more importantly, it shouldn’t have high fluctuations in the cumulative profit graph. If the standard deviation is low, there would be lesser fluctuations in the graph. If the graph shows high fluctuations/volatility, despite the returns being high, the Alpha will not be deemed good enough. An Alpha is considered to be “good” if:

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The graph above for Alpha expression **-Delta(close, 5)** shows several significant drawdowns, as well as a flattening of returns in 2017. The table below marks this Alpha as Inferior (Needs Improvement). PnL and Sharpe for 2017 drop low, and drawdown is large in 2014 and 2015. This Alpha is Inferior (Needs Improvement) due to high volatility and low returns.

```
1  -delta(close, 5)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	2	1	0	Market	On	Off	Verify

IS Summary INFERIOR Stage IS OS

Aggregate Data 2013-2018

Sharpe	Turnover	Fitness	Returns	Margin
1.52	40.32%	0.65	7.39%	3.66‰

Year	Sharpe	Turnover	Fitness	Returns	Margin	Long Count	Short Count
2013	1.47	40.50%	0.52	5.12%	2.53‰	1529	1390
2014	1.45	39.88%	0.70	9.36%	4.69‰	1497	1416
2015	1.18	41.25%	0.48	6.80%	3.30‰	1468	1426
2016	2.78	40.19%	1.51	11.80%	5.87‰	1480	1422
2017	1.00	39.87%	0.29	3.39%	1.70‰	1516	1399
2018	2.42	38.26%	1.38	12.43%	6.50‰	1629	1275

Correlation

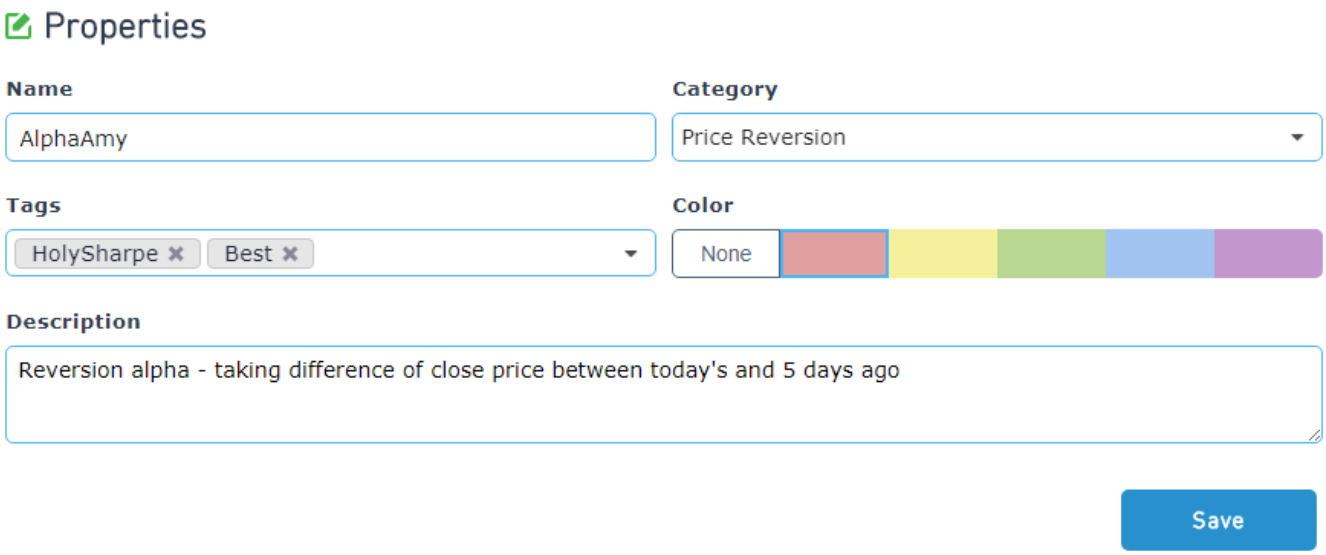
Self Correlation Highest Correlation - Last Run: Wednesday, January 16th 2019, 4:29:51 pm

Use the green refreshing button in the Correlation block to get the information about the correlation of the currently simulated Alpha with the Alphas in your own OS (Out-of-Sample) pool. This will be explained further in the Simulation Results page.

The image below shows the Properties of the Alpha. You can name your Alpha, assign a

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page.



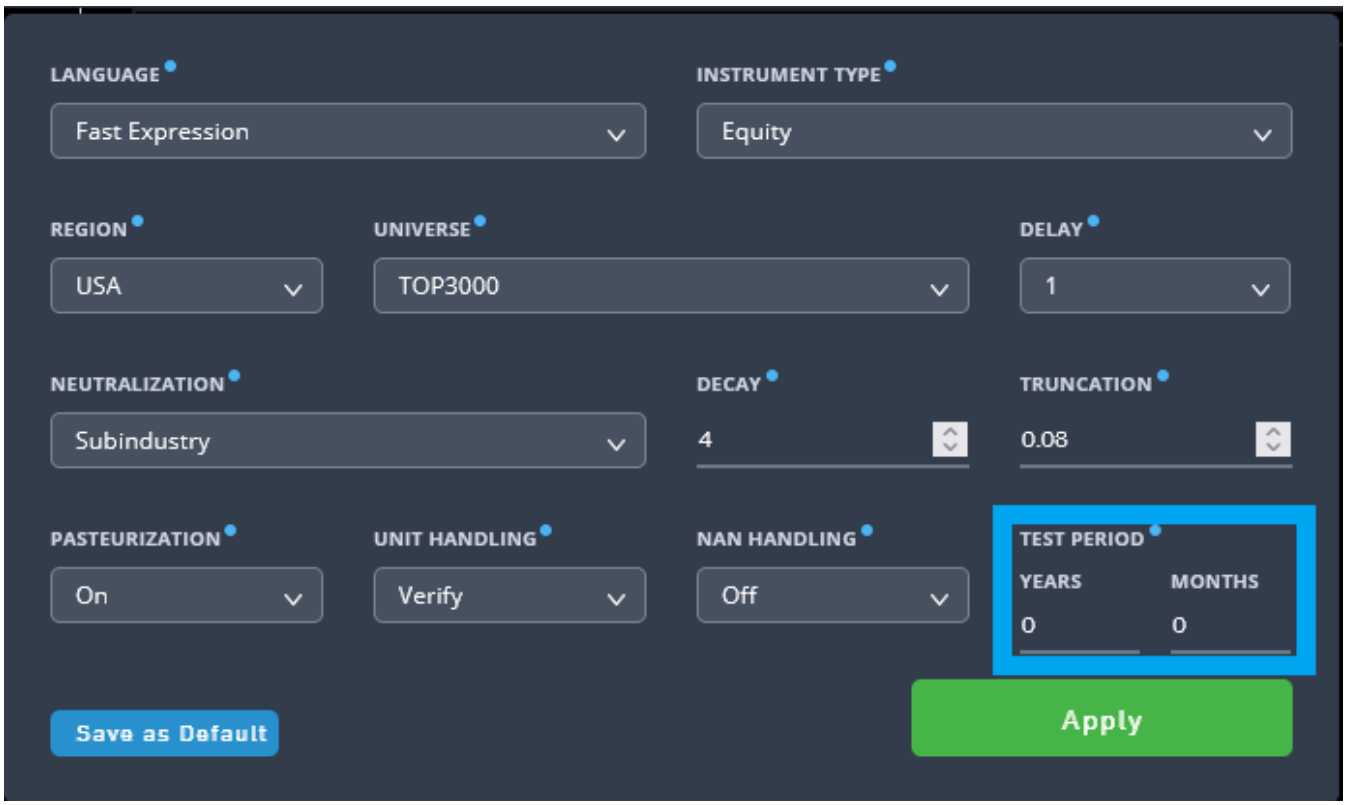
To Submit Alpha for OS Test, click the "Submit Alpha" button in the Submission tab of the results panel. This will check if the Alpha meets the Correlation and Sharpe criteria before submitting it.

Next: Test Period

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Documentation Next: How to choose the Simulation Settings

Test Period



The Test Period is a feature designed to enhance your Alpha and SuperAlpha testing process. This tool allows you to set a separate test period from your IS period, providing a more flexible approach to testing your research ideas.

Using the Feature:

The Test Period feature is designed to help you avoid overfitting. It allows you to divide your In-Sample (IS) period into a Train and Test period. The Train period can be utilized to develop your Alphas and SuperAlphas, while the Test period is ideal for validating them. An Alpha or SuperAlpha that is developed based on the simulation results of Training Period and performs well in both periods is likely a strong candidate for submission and may have avoided overfitting.

While choosing a Test period does not directly affect the simulation, it influences the statistics and the visualization. The submission tests will run on the entire 5-year period, with the simulation running on the entire 5-year IS. However, if a testing period is chosen, the simulation stats will be divided into two sections: one covering the training period and

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1. Selecting the Test Period: In Simulation Settings, you can define a test period corresponding to the final 0-5 years of the IS period. By default, no test period is set (0 years).
2. Visualizing the Test Period: The Stats Summary defaults to the training period. You can view the stats for the test period by clicking on the “Show test period” button.
3. Identifying the Test Period on Graphs: The lines representing the test period on the graphs are colored orange.
4. Choosing the Stats Summary: You can select between the Stats Summary for the test period or the entire IS period by choosing the “TEST” or “IS” in the Summary section, respectively.
5. Hiding the Test Period: A button “Hide test period” allows you to hide the test period, if desired. Note that an Alpha or SuperAlpha can only be submitted when the Test Period is revealed by clicking on the “Show test period” button.
6. Understanding the Stats: The yearly IS stats are divided between Train and Test periods, represented by blue and orange indicators respectively.



A. Orange - test period PnL, Blue - Train period PnL. B. View IS summary by selecting different periods

Prev: Simulate your first Alpha

Next: How to choose the Simulation Settings

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How to choose the Simulation Settings

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1. Language
2. Instrument type
3. Region and Universe
4. Delay
5. Decay
6. Truncation
7. Neutralization
8. Pasteurize
9. Nan Handling
10. Unit Handling

The settings panel can be found by clicking the Settings button at the top right hand corner of the Simulate page. You can specify parameters like language, instrument type, universe, delay, neutralization, etc., which will be applied to your next simulation after clicking the "Apply" button.

Language

Fast Expression is available on BRAIN platform. To learn more, refer to Available Operators

Instrument type

Only Equity instrument type can be used at the moment

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regions Europe and Asia are currently available only to our research consultants.

Universe is a set of trading instruments prepared by BRAIN platform. For example, "US: TOP3000" represents the top 3000 most liquid stocks in the US market (determined by highest average daily dollar volume traded).

Delay

Delay refers to the availability of data, relative to decision time. In other words, delay is the assumption of when we can trade stock once we decide on a position.

Assume that you are looking at the data today before market close and you decide that you want to buy stock. We can choose an aggressive approach and trade stock in the time left till market close. In this case, the position is based on data available on the same day (today). This is called Delay 0 simulation.

Alternatively, we could choose a conservative trading strategy and trade stock the next day(tomorrow). Then the position is achieved tomorrow and it is based on today's data. In this case, there is a lag of 1 day. This is called Delay 1 simulation. In expression language, delay is applied automatically and you do not have to bother about it.

Decay

This performs a linear decay function over the past n days by combining today's value with previous days' decayed value. It performs the following function:

$$Decay_linear(x, n) = \frac{x[date] * n + x[date-1] * (n-1) + \ldots + x[date-n-1]}{n + (n-1) + \ldots + 1}$$

Legal values for Decay: Integer *'n'* where *n >= 0. NOTE: Using negative or non-integer values for Decay will break simulations.*

Tip: Decay can be used to reduce turnover, but decay values that are too large will attenuate the signal.

Truncation

The maximum weight for each stock in the overall portfolio. When it is set to 0, there is no restriction

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outside this range can impact or fail simulations.)

Tip: Truncation aims to guard from being too exposed to movements in individual stocks. The recommended setting is between 0.05 and 0.1 (entailing 5-10%).

Neutralization

Neutralization is an operation used to make our strategy market/industry/sub-industry neutral. When Neutralization = “Market” it does the following operation:

Alpha = Alpha – mean(Alpha)

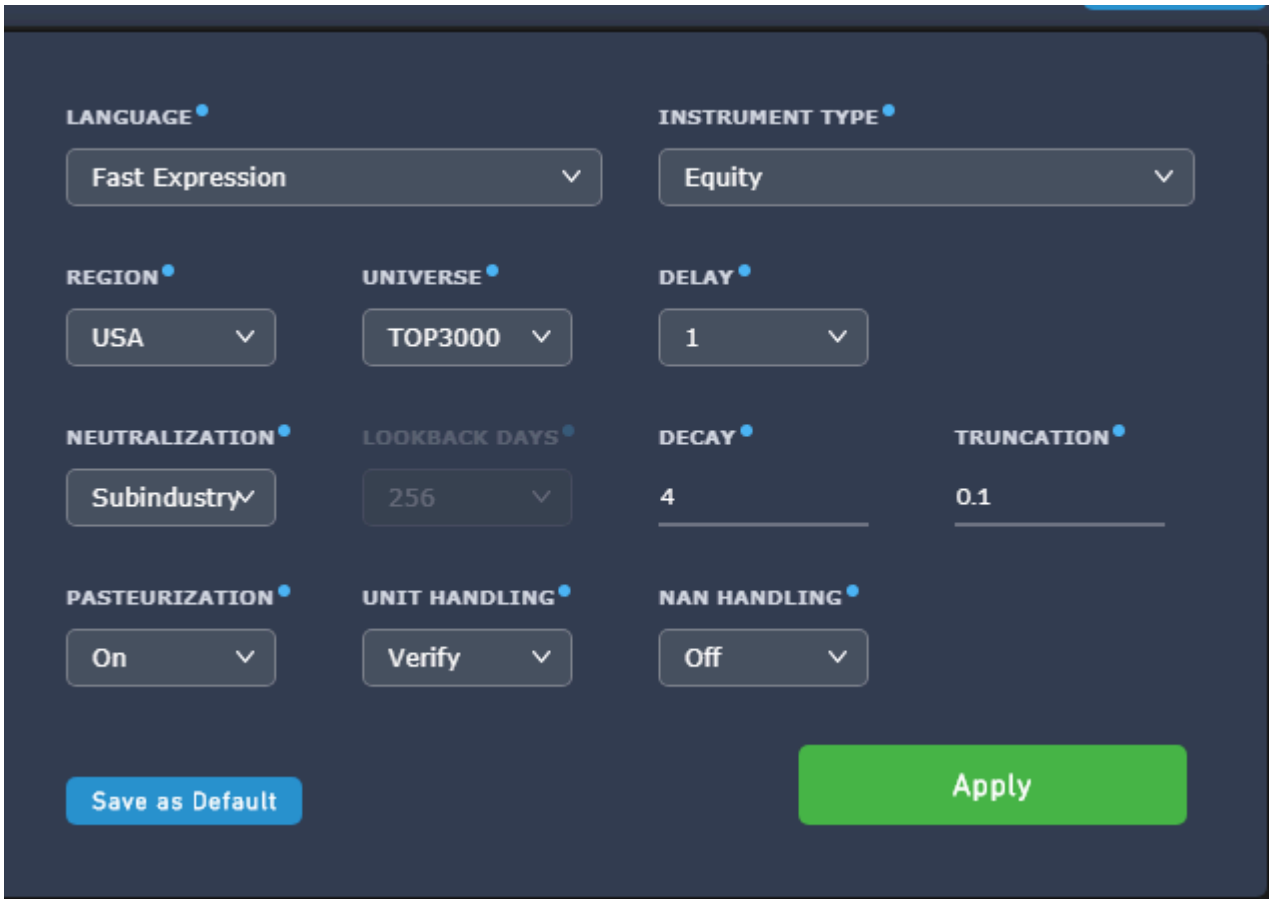
where Alpha is the vector of weights.

Basically, it makes the mean of the Alpha vector zero. Thus no net position is taken with respect to the market. In other words, the long exposure cancels out the short exposure completely, making our strategy market neutral.

When Neutralization = Industry or Subindustry, all the instruments in the Alpha vector are grouped into smaller buckets corresponding to industry or sub-industry and neutralization is applied separately to each of the buckets. For illustration of industry/subindustry classification, see GICS (note: this is not necessarily the same classification standard used by BRAIN platform).

To learn more about Neutralization, refer the Neutralization FAQ section.

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Pasteurize

Pasteurization replaces input values with NaN (pasteurizes) for instruments not in the Alpha universe. When Pasteurize = 'On', inputs to will be converted to NaN for instruments not in the universe selected in Simulation Settings. When Pasteurize = 'Off', this operation does not happen and all available inputs are used.

Pasteurized data has non-NaN values only for instruments in the Alpha universe. While pasteurized data contains less information, it may be more appropriate when considering cross-sectional or group operations. The default Pasteurize setting is 'On'. Researchers can switch it to 'Off' and use pasteurize(x) operator for manual pasteurization.

Example

Assume the following settings are used: Universe TOP500, Pasteurize: 'Off'. The following code calculates the difference between sector rank of sales_growth in Alpha universe and sector rank of sales_growth among all instruments:

```
1  group_rank(pasteurize(sales_g
```

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Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	4	1	0.01	Market	On	Off	Verify

The pasteurize operator in the first group_rank pasteurizes input to the Alpha universe (TOP500), while the second group_rank ranks sales_growth among all instruments.

Nan Handling

NaNHandling replaces NaN values with other values. If NaNHandling: 'On', NaN values are handled based on operator type. For time series operators, if all inputs are NaN, 0 is returned. For group operators returning one value per group (e.g. groupmedian, groupcount), if the input value for an instrument is NaN, the value for the group is returned.

If NaNHandling : 'Off', NaNs are preserved. For time series operators, if all inputs are NaN, NaN is returned. For group operators, if the input value for an instrument is NaN, NaN is returned. Researchers should handle NaNs manually in this case. The default setting NaNHandling value is 'Off'. Some ways to manually handle NaN values can replicate "On" behavior.

Example

```
ts_zscore(etz_eps, 252)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	4	1	0.01	Market	On	Off	Verify

Assume NaNHandling = 'On'. Then for a stock with etz_eps == NaN for all 252 days, 0 is returned. However, ts_zscore(x, d) also returns 0 when x == tsmean(x, d), which is different from x == NaN ("no data is available"). This means that NaNHandling = 'On' increases coverage, but may introduce ambiguous information into the Alpha.

If NaNHandling = 'Off', NaNs can be handled other ways:

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Example

```
groupmax(sales, industry)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	3	1	0.01	None	On	Off	Verify

When NaNHandling = 'Off' and sales is NaN for a given instrument, the operator's output is NaN. When NaNHandling = 'On' and sales is NaN for a given instrument, the operator's output is the maximum value of sales in the instrument's industry.

Unit Handling

Unit Handling option allows raising a warning when incompatible units are used in an operator. This warning appears if expression uses data fields that are incompatible, for example, a warning will be shown for an attempt to add price to volume.

Prev: Test Period

Next: ⭐ How BRAIN platform works

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Documentation Next: How to view your Alphas?

⭐ How BRAIN platform works

The below post illustrates in detail how the BRAIN platform works and what happens in the background when you simulate an alpha. Even though you'll never need to do these calculations yourselves, developing an intuition for them will help you in the alpha making process.

Imagine market data being a matrix, with each row representing one date and each column representing one stock. For example, the matrix for close price data of stocks in universe US TOP3000 would look like this:

Dates	MSFT	HOG	AAPL	GOOG	PG	...
20100104	30.95	25.46	214.01	626.75	61.12	...
20100105	30.96	25.65	214.38	623.99	61.14	...
20100106	30.77	25.59	210.97	608.26	60.85	...
20100107	30.452	25.8	210.58	594.1	60.52	...
20100108	30.66	25.53	211.98	602.02	60.44	...

When you input the simulation settings and click "Simulate", the BRAIN platform will evaluate the Alpha expression against the matrix of market data for each date in a 5 year span, taking a long or short position for each financial instrument to generate the PnL chart.

When you click "Simulate":



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generated.

Normally, in an alpha simulation, there would be between 200 and 3,000 stock instruments in the universe. But to better understand this concept, we'll assume a hypothetical scenario in which the simulation universe has only eight stocks. We simulate the expression **rank(-returns)** with market neutralization, Delay 1 and Decay 0 settings for now.

The hypothesis in this expression is that we want to buy, or go long on, those stocks tomorrow that had negative or comparatively lower returns today, and we want to sell, or go short on, those stocks tomorrow that had positive or comparatively higher returns today.

We've used the rank operator, which ranks the input values inside the operator and return values equally distributed between zero and 1. This is an example of a reversion idea.



In Column B, we have the eight stocks in the alpha vector. Column C shows the returns of these stocks as of February 1st. These serve as the input data of the alpha expression.

Step1: Evaluate the expression for each stock to generate the alpha vector for the given date.

In our case, this date would be February 2nd, because we've assumed Delay 1 settings. The Delay 1 setting uses data as of T-1 date to create the alpha vector as of T date.

To produce the alpha vector, the simulator performs the rank operation on negative returns and produces a vector of values corresponding to each stock.

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The resulting vector depends on the operators used in the alpha expression. In our case, since we've used the rank operator, we see equally distributed values between 0 and 1 in Column D. Note that the stock with the lowest return has the highest value, and vice versa, in line with our hypothesis.

***Step 2**: From each value in the vector, subtract the average of the vector values in the group. Sum of all vector values = 0.* This is called neutralization.

The group can be the entire market, but we can also perform this neutralization operation on sector, industry or subindustry groupings of stocks.

	B	C	D	E	F	G	H	I	J	K	L
1	How does the Expression become a PnL curve										
2	Stocks in the Alpha vector	Stock Returns on 1-Feb	Alpha value on 2-Feb : rank(-returns)	Subtract Average	Centred around 0	Absolute value	Normalized weights on 2-Feb	Absolute value	Allocate $ 20 Mn capital (Position on 2nd Feb)	Suppose Returns on 2 Feb are	$Mn profits on 2 Feb would be
3	Stock 1	3.0%	0.00	(0.50)	-0.5	0.50	-0.22	0.22	-4.4	2.0%	-0.09
4	Stock 2	2.5%	0.14	(0.50)	-0.4	0.36	-0.16	0.16	-3.1	2.0%	-0.06
5	Stock 3	1.8%	0.29	(0.50)	-0.2	0.21	-0.09	0.09	-1.9	-2.1%	0.04
6	Stock 4	0.6%	0.43	(0.50)	-0.1	0.07	-0.03	0.03	-0.6	-2.1%	0.01
7	Stock 5	0.2%	0.57	(0.50)	0.1	0.07	0.03	0.03	0.6	1.0%	0.01
8	Stock 6	-0.9%	0.71	(0.50)	0.2	0.21	0.09	0.09	1.9	0.0%	0.00

Since we have only eight stocks in our simulation universe, we've assumed to neutralize the stocks over the market.

So we take the average of the numbers in Cell D12 and subtract the average from each stock. This gives us a new vector in Column F. Note that both the sum and the average of these new numbers are now zero. Also, the sum of positive values is equal to the sum of negative values.

***Step 3:** The resulting values are scaled or 'normalized' such that absolute sum of the alpha vector values is 1. These values can be called as normalized weights.*

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That means, we sum the absolute values of each row and find the sum, which is 2.3. Then we divide each row by this sum, which results in normalized values. By normalize, we mean that the total absolute sum of Column H is 1. We can also call this vector a normalized vector of weights.

Note: On each iteration/day, the expression rank(-returns) will have access to all the data for returns up to that day, and the matrix will grow by one line every day until it reaches the most recent date. The role of the expression is to transform the input matrix to an output vector of weights as we see in this hypothetical example.

Step 4: *Using normalized weights, the BRAIN simulator allocates capital (from a fictitious book of $20 million) to each stock to construct a portfolio.*

Column J has a total of $20 million of fictional money allocated to the stocks, using the normalized weights in Column H. This means we have a position of minus $4.4 million in Stock 1 — that is, we've shorted $4.4 million worth of Stock 1 — and a long position of $0.6 million in Stock 5. That is, we've invested $0.6 million in Stock 5.

This is called long-short market neutralization, and it's the backbone of creating these predictive models, or alphas, on BRAIN. With this technique, a strategy can have the potential to be profitable regardless of the direction of the market.

Step 5: *Calculate next day PnL generated by the alpha based on observed stock returns*

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each stock, based on the returns each stock had that day.

Suppose the actual returns on these stocks as of February 2nd are as shown in Column K. We see that although we expected Stock 1 and Stock 2 to fall in price, they actually went up, so we had a loss, shown in Column L.

We expected Stock 6 to go up in price, but it stayed flat. So we were wrong about three stocks, but we were right about five. In total, we made a gain of $0.03 million on this day with our alpha, calculated by adding the PnLs of all stocks in our vector. This is how the simulator calculates the PnL generated by the alpha for any given date.

Step 6: *Perform the operations in Step 1 to Step 5 for each date in a several-year history span also called the In-sample period (IS) to get daily PnL generated for each day*

For each day, the expression is evaluated and the values in the Alpha output vector represent the weights to allocate to each stock. Alpha weights are not how much you want to buy or sell, but a weighting position you would reach this day. These weights are multiplied by book size (total money invested in the portfolio) to get the dollar value held in each stock. For example, if the Alpha weight (after neutralization and scaling) for MSFT is 0.2423, then we'll have MSFT stocks with the total value 0.2423*book size.

The weight can be negative, meaning you would take a short position on these stocks. If the value is positive, you would take a long position on these stocks, i.e. buy the stocks. A NAN value would mean no weight is allocated to that instrument (i.e. no money is allocated). The value of stocks you buy/sell on a particular day is determined by the difference between weights today and weights yesterday. The percentage of your portfolio traded in a day (by dollar value) is called 'turnover'. The turnover reported in simulation results is the average daily turnover over the simulation.

Step 7: *Calculate the cumulative PnL of the alpha from the start of the in-sample period to get the PnL chart of the alpha.*

Based on those daily positions, PnL is calculated and displayed. By default, the BRAIN

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collection of securities.)

This can be better understood with the help of the PnL chart of the alpha in our example rank(-returns)



In this chart, we have an IS period of five years, from February 2016 to January 2021. Using the steps we discussed in our example, the simulator would calculate the daily PnL of the alpha and derive the cumulative PnL chart, as we see here. Note that the two years from February 2021 to January 2023 are not visible to us in the simulation window. That's called the out-of-sample, or the OS, period. After you submit an alpha, several tests are run to analyze the alpha's performance in the OS period. An alpha that passes both the in-sample and out-of-sample tests can be said to be a robust alpha.

This is how the BRAIN simulator creates the PnL chart from an alpha.

In our example, we've assumed that we're using market neutralization and Decay 0 settings. But if we used any other neutralization settings, the same operations would be performed on the alpha.

Say we have 80 stocks in our simulation universe — ten industries with eight stocks each. The simulator would perform the same operations (first Step 1 to Step 5) on each of the ten groups and finally add the PnL from each group to get the daily PnL of the alpha and create the cumulative PnL chart (Step 6 and Step 7)

However, if we introduce decay into our alpha settings, an additional step must be performed to get the final alpha vector.

Suppose we use a decay of 3 in our simulation settings. The final vector of weights in the alpha would be calculated by combining today's value with the previous day's decayed value. In our example, we calculated the normalized weights in the alpha as of February 2nd. Let's assume that the normalized weights of stocks in the alpha vector on February 1st and January 31st are as shown in Columns N and O, respectively

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Then the final weights in the alpha would be calculated using the given weighted average formula:

$$Decay_linear(x, n) = \frac{x[date] * n + x[date-1] * (n-1) + \ldots + x[date-n-1]}{n + (n-1) + \ldots + 1}$$

which is implemented in Column P. Using this new derived vector, the simulator would calculate the daily PnL and consequently the cumulative PnL chart. Note that even if decay is used, more weight is assigned to the most recent values. So decay is a very important factor in reducing transaction costs or turnover, as it includes information from previous days, preventing the alpha from being very reactive.

To summarize, once we input the alpha expression and simulation settings in the BRAIN simulator, it performs the operations discussed above to take long or short positions for each financial instrument and generates the PnL chart.

Prev: How to choose the Simulation Settings

Next: How to view your Alphas?

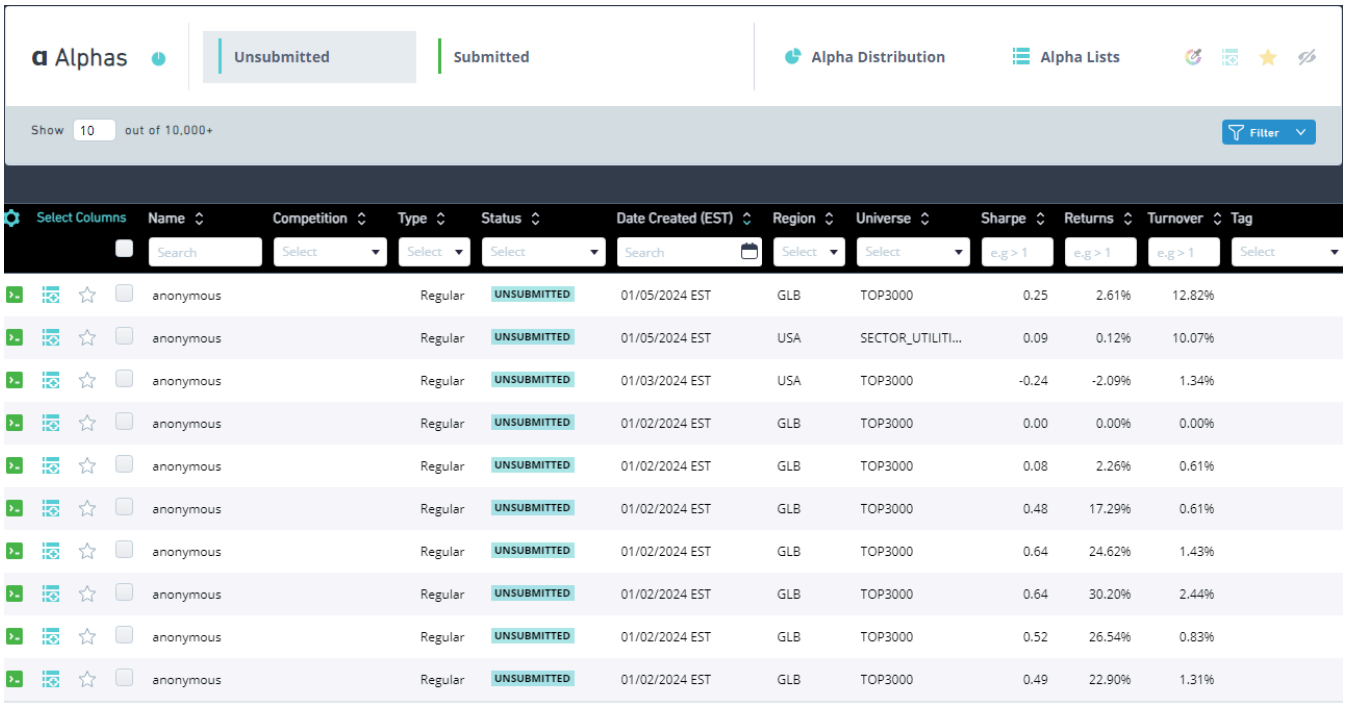
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How to view your Alphas?

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1. Distribution of Active Alphas



The "Alphas" page shows a summary of the Alphas simulated so far. The controls available on this page are:

- **Stage:** The Alphas are grouped based on the stage - Unsubmitted and Submitted.
- **Favorite**: Alphas can be marked as favorite by clicking the



The same procedure on favorite Alphas allows to unfavorite an Alpha.

- **Hidden**: Alphas can be hidden by selecting their checkbox and clicking the

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- **Columns, Filters and Sorting:** Columns can be added or removed in the Alphas page from 4 categories – **Summary, Settings, Performance, Properties**

Filters can be used to view a subset of all Alphas. The parameters by which you can filter are: **Name, Category, Code, Language, Color, Date Created, Decay, Drawdown, Favorite, Fitness, Hidden, Margin, Neutralization, PnL, Region, Returns, Sharpe, Status, Tags, Truncation, Turnover and Universe.** *[Note: Margin is the PnL divided by dollars traded. It is the same as the "Profit per $ traded" column on the results page. The unit bpm stands for basis points (margin), where each basis point represents 1/100th of 1% margin. I.e. 500 bpm would represent 5% margin.]*

Fields to be sorted and the order of sorting can be chosen in this pop-up or by clicking on the column header to cycle between increasing and decreasing order of sorting. The number of Alphas to be displayed per page can also be adjusted in this section.

- **View Code / Simulation:** Click the



icon to view the code for an expression. You can then click Alpha name to open Alpha information in separate block, and use "Clone Alpha" button to open this Alpha in simulation window.

- **Alpha Lists:** To compare the performance of two or more Alphas, first add the Alphas to the list by selecting the Alphas and clicking the



Distribution of Active Alphas

This page visually represents the distribution of your submitted Alphas across three key dimensions: Region, Delay, and Data Category. This graphical visualization is divided into three layers, each representing one of these dimensions. The number displayed on each segment, visible upon hovering your mouse over it, indicates the number of Alphas in that subset and their proportion within the larger set.

Here's how to navigate this feature:

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dive deeper.

2. Upon selecting a region, you'll move to the second layer representing Delay. Here, you can choose between the available options.
3. Clicking on a delay segment will lead you to the third layer, which shows the Data Category distribution for your submitted Alphas, specific to the chosen Region and Delay.

This feature not only provides an overview of your submitted Alphas but also guides you in diversifying your research. Here's how:

- Avoid submitting more than 50% of your Alphas in a single region.

- Try to submit at least 5% of your Alphas with Delay-0.

- Avoid submitting more than 30% of your Alphas in any single intersection of Region, Delay, and Data Category.

This feature provides a color-coded visual cue to highlight the concentration of Alphas in a specific intersection of Region, Delay, and Data Category. The color gradient ranges from blue to red, with red indicating high concentration.

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Next: ⭐ Alpha Examples for Beginners

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Documentation

⭐ Alpha Examples for Beginners

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1. EBIT vs CAPEX
2. Future Investment and Dividend
3. Room for growth
4. Pay back the debt
5. Retained earnings
6. Pretax Income

EBIT vs CAPEX

Hypothesis

Stocks with higher EBIT compared to CapEx can be a sign of the company not investing much in growth and the stock may not grow as much, thus we should sell those stocks.

Implementation

CapEx (capital expenditure) is money spent on acquiring/maintaining fixed assets. Compare CapEx with EBIT, which is a company's net income before interest and taxes, and short stocks with inadequate CapEx.

Hint to improve the Alpha

Can different neutralization settings improve the alpha performance?

```
-rank(ebit/capex)
```

Open example alpha in Simulate

Simulation Settings

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		Simulate		Alphas		Data	Competitions (2)		m
USA	TOP3000	Fast Expression	0	1	0.01	Sector	On	Off	Verify

Future Investment and Dividend

Hypothesis

Retained Earnings is a measure of a firm's net income that it has accrued over time and saved after the distribution of dividend. So Retained Earnings are an indicator of the capability of future investment and dividend.

Implementation

Use sharesout to normalize the amount of retained earnings; then apply ts_delta() operator to capture the change of the ratio over the last three months (one quarter). Finally use rank() operator to normalize the result.

Hint to improve the Alpha

Think about how many trading days in one quarter. Maybe there are less than 90 trading days in three months.

```
1  rank(ts_delta(retained_earnin
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
CHN	TOP3000	Fast Expression	5	1	0.01	Subindustry	On	Off	Verify

Room for growth

Hypothesis

Companies with high asset fair value but comparatively lower EBIT are likely companies that have invested for growth over the past years and have more room for growth in the future

Implementation

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signal for companies with lower asset fair value.

Hint to improve the Alpha

Can the alpha work well if comparing among companies that are already more sizeable (TOP500), because companies that are too small may consider this metric as too risky as it is a sign of lower cashflows.

```
alpha = -group_rank(fnd2_ebit

group_rank(fn_assets_fair_val
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	0	1	0.01	Subindustry	On	Off	Verify

Pay back the debt

Hypothesis

It is usually safer to go long on companies that can easily pay back the short term debt using high liquid assets.

Implementation

Zscore of the ratio between cash and short term debt is calculated with higher readings refer to higher ratio when compared with the market.

Hint to improve the Alpha

Try comparing a stock with its peers instead of the whole market.

```
zscore(cash_st/debt_st)
```

Open example alpha in Simulate

Simulation Settings

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		Simulate		Alphas		Data	Competitions (2)		m
USA	TOP500	Fast Expression	0	1	0.01	Subindustry	On	Off	Verify

Retained earnings

Hypothesis

Retained earnings are the cumulative net earnings of a company after dividend payments to the shareholders. In certain situations, shareholders might want to receive their dividends and realize their profit. Thus, some shareholders might not be happy with excessive retained earnings.

Implementation

We should long the stock that has its retained earnings decreasing and vice-versa.

Hint to improve the Alpha

Can different neutralization setting improve the alpha performance?

```
-ts_rank(retained_earnings,25
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
USA	TOP3000	Fast Expression	10	1	0.01	Sector	On	Off	Verify

Pretax Income

Hypothesis

Pretax income is a good measure of a company's financial health and profitability. Companies with increasing pretax income may have good growth prospects and higher potential for increasing stock price. Thus, we would long stocks with increasing pretax income and short stocks with decreasing pretax income.

Implementation

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Boost the signal with sales data. For example, if the company has higher sales, it is more likely to outperform.

```
1  quantile(ts_rank(pretax_incom
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit I
CHN	TOP3000	Fast Expression	4	1	0.01	Market	On	Off	Verify

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Clear these tests before submitting an Alpha

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2. Alphas in CHN region
3. Tests on SECTOR_UTILITIES_TOP3000 Universe of USA Region
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5. Weight
6. Sub Universe Test
7. Interpreting Status Messages in Simulation Results
8. Selecting Alphas For Submission

- The following performance tests are run till the end of the In-Sample Period.
- The 'Submit Alpha' button (in the Submission tab of the simulation results panel) is used to start out-of-sample (OS) testing for Alphas meeting the performance and correlation cutoffs
- **Only submitted Alphas are considered for scoring.** Submitted Alphas show up on the Out-of-Sample tab of the Alphas page.

Below table is for submission test for Alphas excluding CHN region

Tests on alphas excluding CHN region

SUBMISSION CRITERIA	THRESHOLDS FOR USERS
Fitness	At least “Average”: Greater than 1.3 for Delay-0 or Greater than 1 for Delay-1
Sharpe	Greater than 2 for Delay-0 Alphas or Greater than 1.25 for Delay-1 Alphas
Turnover	1% < Turnover < 70%
Weight test	Max weight in any stock < 10%. This measures if sufficient

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	universe size. You can find detailed example below.
Self Correlation	<0.7 PNL correlation. Or Sharpe at least 10% greater than other correlated alphas submitted by user

Alphas in CHN region

- The China market has a high cost of trading, thus requiring higher returns than other regions. Thus the submission criteria are **Sharpe >= 1.625, Returns >= 6.3% and Fitness >= 1.0 for D1; Sharpe >= 2.6, Returns >= 8.9% and Fitness >= 1.3 for D0**
- Apart from usual robustness tests such as sub universes, turnover, fitness and weight, there is an additional test exclusive to the China research region: **Robust universe test performance**: Alphas are considered good if the robust universe component retains at least 40% of the returns and Sharpe of the submission version.

Tests on SECTOR_UTILITIES_TOP3000 Universe of USA Region

SUBMISSION CRITERIA	REQUIREMENTS
Sharpe	Greater than 1 for both Delay- 0 and Delay- 1 Alphas
Turnover	1%<Turnover<70%
Weight Test	Max Weight in any stock <10%
Fitness and IS Ladder Tests	Not Mandatory, but suboptimal value triggers a warning note

Self-correlation

- Alphas can also qualify if their Sharpe is greater, by 10% or more, than that of all Alphas with which their correlation is higher than the cutoff.
    - For example, if your earlier submitted Alpha X has a Sharpe of 3.18, you can submit a highly correlated Alpha Y, if its sharpe is 3.5 or more
    - This allows for making improvements to an existing Alpha.
    - The Sharpe value used for this comparison (3.18) is visible in the correlation

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operates on the intersect of the selected Alpha's PnL time periods.

Weight

Alphas are also tested on the distribution of Alpha weights across stocks. Alphas can fail this test if:

- Too few stocks are assigned weight for significant number of days in a year. Note that assigning zero weights to all stocks at the start of the simulation does not fail this condition, it only applies after the Alpha starts assigning weights. The exact number of minimum stocks varies with the simulation universe.
- Alpha weight is too concentrated in any one stock. For example, if one stock has 30 percent of all Alpha weight, it will fail this test.

Sub Universe Test

subuniverse_sharpe >= 0.75 * sqrt(subuniverse_size / alpha_universe_size) * alpha_sharpe

Interpreting Status Messages in Simulation Results

When "Check Submission" or "Submit Alpha" button is pressed, tests are performed in the order described below. In case Alpha fails any of the tests, respective Test Message is displayed.

SR. NO.	TEST RESULT	TEST MESSAGE
1	Alpha fails Weight test	Maximum weight on an instrument is greater than 10% OR Weight is too strongly concentrated or too few instruments are assigned weight.
2	Alpha fails Correlation test	Reduce max correlation
3	Alpha fails fitness test	Improve fitness
4	Delay 0 Alpha fails checkDelay1Sharpe	Alpha better suited for Delay 1
5	Alpha fails SubUniverse test	Improve Sharpe in SubUniverse

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would be "Improve Sharpe or reduce turnover"

Selecting Alphas For Submission

- Do not submit Alphas as soon they clear the performance cutoff. Improve the idea until you have the best version: in terms of both performance and correlation.
- However, do not spend extraordinary amount of time improving a single idea either:
  - It is generally better to try out new ideas with low correlation to previous ones than to improve performance of Alphas with high correlation.
  - Generally, low correlation is more important than minor increase in performance. Example: An Alpha with slightly better performance but high correlation is worse that an Alpha with slightly lower performance but much lower correlation

Next: Parameters in the Simulation results

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Parameters in the Simulation results

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1. Return
2. Sharpe and IR
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5. IS Summary
6. Self Correlation

In the Simulation result page, you will find a ratings panel in the Stats tab of Results that says Spectacular, Excellent, Good, Average or Needs Improvement depending on your Alpha's Fitness as shown below:

Label	Fitness for Delay 1	Fitness for Delay 0
Spectacular	> 2.5	> 3.25
Excellent	> 2	> 2.6
Good	> 1.5	> 1.95
Average	> 1	> 1.3
Needs Improvement	<= 1	<= 1.3

Return

Return is the gain or loss of a security or portfolio in a particular period. Return consists of the income received plus capital gains, relative to the amount of the investment. In BRAIN paltform, return = annualized PnL / half of book size.

Sharpe and IR

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defined as the ratio of a portfolio's mean daily returns to the volatility of those returns.

$$IR = \frac{mean(PnL)}{stdev(PnL)}$$

where PnL is the daily profit and loss, in dollars.

Sharpe is the annualized version of the IR statistic, i.e. Sharpe = sqrt (252)*IR ≈ 15.8*IR; where 252 is the average number of trading days (days the markets are open) in the USA in a year.

Sharpe or IR measures the returns of an Alpha while attempting to identify its consistency. The higher the IR, the more consistent the Alpha's returns are, and consistency is an ideal trait. High Sharpe (or IR) is more desirable than just high return.

Note: Sharpe and IR may be defined somewhat differently elsewhere than in BRAIN platform *.*

Fitness

Fitness of an Alpha is a function of Returns, Turnover and Sharpe:

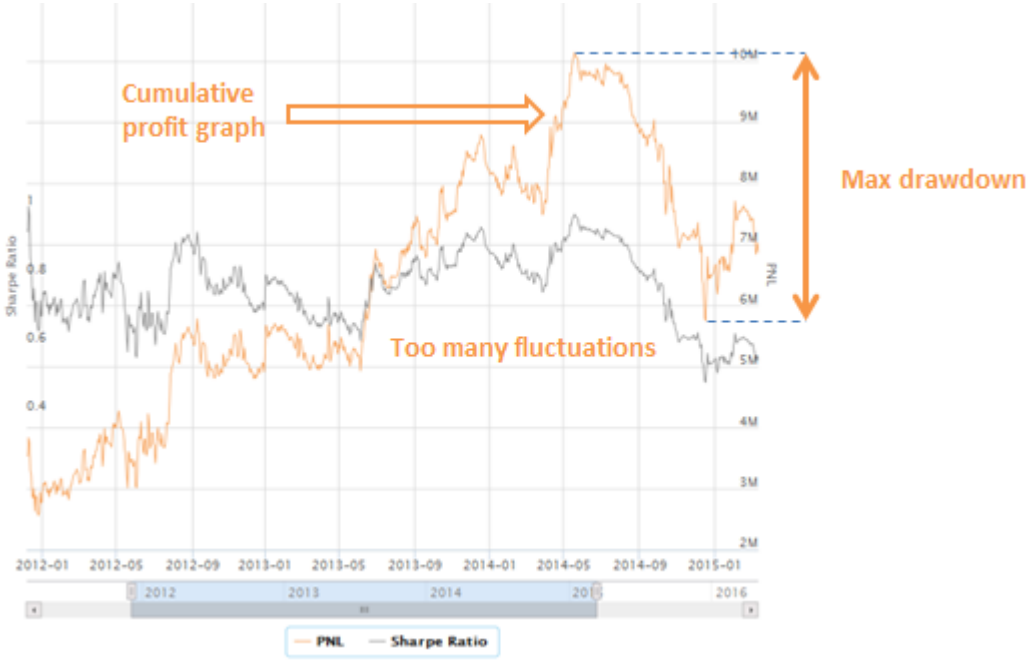
$$Fitness = Sharpe \cdot \sqrt{\frac{abs(Returns)}{max(Turnover, 0.125)}}$$

Good Alphas have high fitness. You can optimize the performance of your Alphas by **increasing Sharpe (or returns) and reducing turnover**. Improving one factor normally has an adverse impact on the other factor. As you work on optimizing your Alpha, an improvement in its fitness is an indication that your changes are having a positive impact.

Cumulative PnL Chart

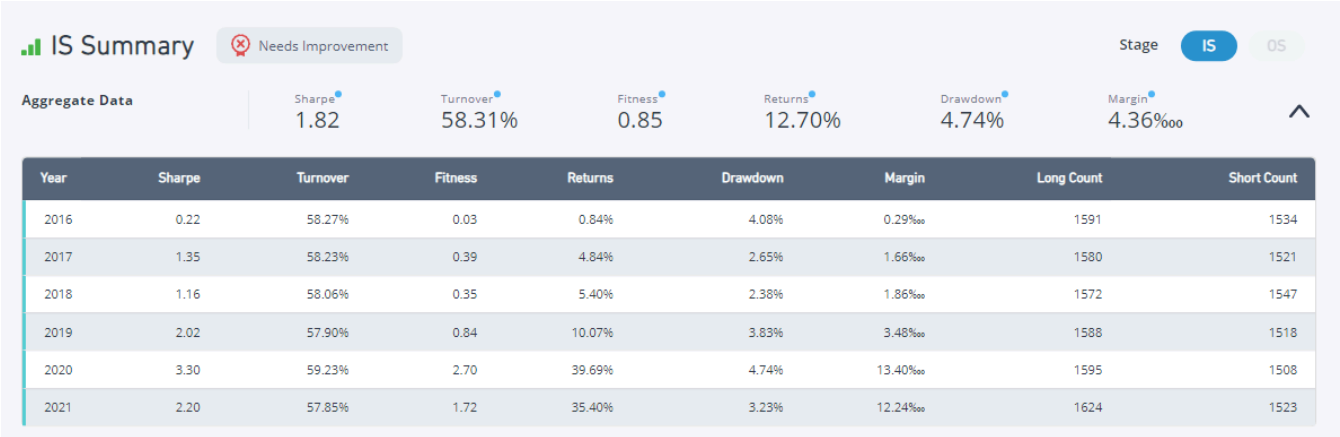
Cumulative PnL Chart: A graph (shown below) of an Alpha's performance (PnL) over entire simulation. This graph can be zoomed in by clicking and dragging below the plot area. Start and end dates for PnL plotting can also be changed here. Clicking the Sharpe Ratio in dropdown menu at the upper right from PnL graph displays the Sharpe ratio graph (Sharpe over time). Make sure that the PnL graph has an upward trend, the Sharpe is high and the Drawdown is kept to a minimum.

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IS Summary

IS Summary: Scrolling down to the Stats block (shown below) of the simulation results shows various metrics about the Alpha's performance.



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2016	0.22	58.27%	0.03	0.84%	4.08%	0.29‰	1591	1534
2017	1.35	58.23%	0.39	4.84%	2.65%	1.66‰	1580	1521
2018	1.16	58.06%	0.35	5.40%	2.38%	1.86‰	1572	1547
2019	2.02	57.90%	0.84	10.07%	3.83%	3.48‰	1588	1518
2020	3.30	59.23%	2.70	39.69%	4.74%	13.40‰	1595	1508
2021	2.20	57.85%	1.72	35.40%	3.23%	12.24‰	1624	1523

Year: The year on which the data was simulated. The last row shows the Alpha's performance over all years.

Long/Short Count: The number of instruments in long or short positions, respectively.

Sharpe: Sharpe = IR * Sqrt(252), where IR = Avg(PnL)/Std_dev(PnL) over the observed time period.

Fitness is defined in the Alpha Performance help page: Fitness = Sharpe *

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size. It signifies the amount you made or lost during the period observed and is expressed in %. Booksize refers to the amount of capital (money) used to trade during the simulation. Booksize is constant and is set to $20 million every day throughout the simulation. Profit is not reinvested, and losses are replaced by cash injection into the portfolio. BRAIN platform assumes you have $10 million and will invest in assets up to $20 million. This is called leverage. Performance (like Returns, Sharpe) is computed on a base of $10 million.

Turnover: Turnover signifies how often one trades. It can be defined as the ratio of value traded to book size. Daily Turnover = Dollar trading volume/Booksize. Good Alphas have low turnover, since low turnover means lower transaction costs.

Margin: The profit per dollar traded; calculated as PnL divided by total dollars traded for a given time period.

PnL: Profit and Loss (PnL) is the money that the positions and trades generate (which means it is the amount of money you lost or made during the year), expressed in dollars.

daily_PnL = sum of (size of position * daily_return) for all instruments, where the daily return per instrument = (today's close / yesterday's close) – 1.0.

Drawdown - the largest reduction in PnL during a time period, expressed as a percentage. It is calculated as follows: find the largest peak to trough drawdown in PnL, and divide its dollar amount by half of book size ($1

Self Correlation

Generate Self Correlation: Clicking the Down Arrow button in a Self Correlation row will produce a table with the performance statistics of up to the 5 most correlated Alphas you submitted that qualified for OS testing. This information is meant to help the user ensure they have a diverse set of Alphas. This information can also be accessed by clicking on the Alpha in the Alphas page.

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Self Correlation

Highest Correlation

0.991

Last Run: Tuesday, January 15th 2019, 10:21:21 am

Name	Instrument Type	Region	Universe	Correlation	Sharpe	Returns	Turnover	Fitness
anonymous (current)	EQUITY	USA	TOP3000	1	1.45	0.026	0.046	1.08
anonymous	EQUITY	USA	TOP3000	0.991	1.39	0.025	0.043	1.05
anonymous	EQUITY	USA	TOP3000	0.407	1.73	0.031	0.044	1.45
anonymous	EQUITY	USA	TOP3000	0.301	1.41	0.025	0.032	1.24
anonymous	EQUITY	USA	TOP3000	0.276	1.28	0.027	0.024	1.36
anonymous	EQUITY	USA	TOP2000	0.244	1.27	0.034	0.037	1.23

Prev: Clear these tests before submitting an Alpha

Next: IS, Semi-OS, and OS

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Documentation

IS, Semi-OS, and OS

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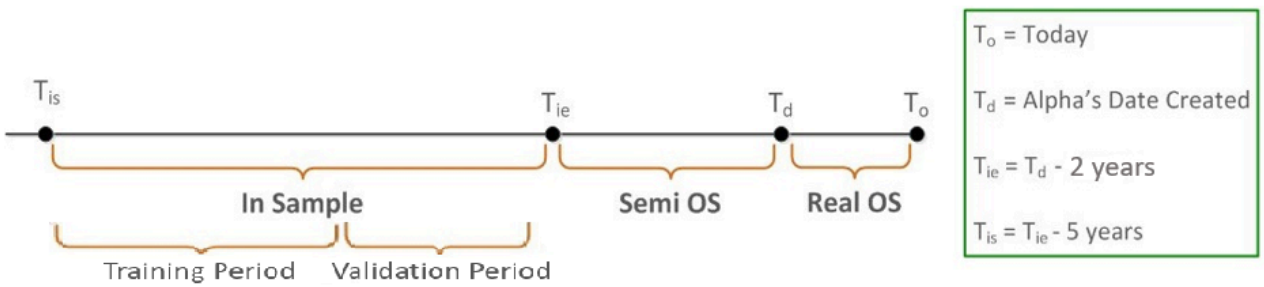
1. IS, Semi-OS & OS
2. Alpha Statuses

IS, Semi-OS & OS

The rolling 5-year In-Sample simulation period begins 7 years ago and ends 2 years ago, updating daily. Using simulation settings, you can divide your In-Sample (IS) period into a Train and Test period. The Train period can be utilized to develop your Alphas and SuperAlphas, while the Test period is ideal for validating them. An Alpha developed based on the simulation results of Training Period and performs well in both periods is likely a strong candidate for submission and may have avoided overfitting.

The latest 2 years of data, the Semi-OS, are hidden for scoring and testing purposes. Consultants have access to a 10-year In-sample period, instead of 5-year.

Keeping the last 2 years of data hidden leads to higher confidence in the Out-Sample (OS) performance of Alphas and their scores. Statistics shown in the OS Tab of **My Alphas** page will be populated as data becomes available by each passing day.



Alpha Statuses

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- Upon submission, the Alpha is assigned the "ACTIVE" status.
- For consultants, ACTIVE Alphas are qualified to accumulate weight and are eligible to contribute to the consultants' quarterly payments, as further described in their respective consulting or service agreements. This ACTIVE status will be kept until the dataset they rely upon is decommissioned or if WorldQuant otherwise decommissions an Alpha, in its discretion.
- In case the dataset in no longer available or there is prolonged underperformance of the Alpha in the Out-Sample period, the Alpha's status is revised to "DECOMMISSIONED".
- Decommissioned Alphas do not accrue weight and are not eligible to contribute to your quarterly payment.



Prev: Parameters in the Simulation results

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